Operating instructions

V 3.0 English

QiTech Winder

QiTech Industries GmbH
Landwehrstraße 55
64293 Darmstadt

EC Declaration of Conformity

According to 2006/42/EC, Annex II, No. 1 A

The content of the EC Declaration of Conformity corresponds to Annex II, No. 1 A of Directive 2006/42/EC.

Manufacturer of the machine:

QiTech Industries GmbH
Landwehrstraße 55
64293 Darmstadt

Representative of the manufacturer:

Milan von dem Bussche
Landwehrstraße 55
64293 Darmstadt

Person authorised to compile the technical file:

Milan von dem Bussche
Landwehrstraße 55
64293 Darmstadt

General designation	Filament winding machine
Type	Filament winding machine
Function	Winding and rewinding of filaments
Model	QiTech Winder / QiTech Winder XL
Trade name	QiTech Winder / QiTech Winder XL
Serial number	

We hereby declare that the above-mentioned machine is in conformity with all relevant provisions of the EC Machinery Directive 2006/42/EC. The machine is also in compliance with all relevant provisions of the following EC Directives:

- 2014/30/EU Electromagnetic Compatibility / European EMC Directive
- 2014/35/EU Low Voltage Directive (protection objectives only)
- 2011/65/EU ROHS Directives (ROHS 2)

The following harmonized standards (or parts of these standards) have been applied:

- DIN EN ISO 12100
- DIN EN ISO 20607

Darmstadt, 01.09.2026



Milan von dem Bussche

Read before commissioning!

These operating instructions are intended to enable all people working on the machine to carry out their work safely and correctly. Every person who operates, maintains or repairs the machine must have read the operating instructions, particularly the safety regulations, before commissioning. These operating instructions must always be available on the machine. The operator must ensure that all information necessary for the safe operation of the machine is made available to the operating personnel. If the operating instructions are available in electronic form, the operator must ensure that the operating personnel are able to view the operating instructions. The original version of these operating instructions is written in English. We ask you to read through the relevant instructions carefully and to observe them thoroughly. After unpacking the machine, check for any damage that may have occurred during transport.

NOTICE: Before using the machine for the first time, please inform yourself about the national regulations (e.g. occupational safety, environmental restrictions). Reproduction and reprinting in any form, even in extracts, requires the written authorization of QiTech Industries GmbH.

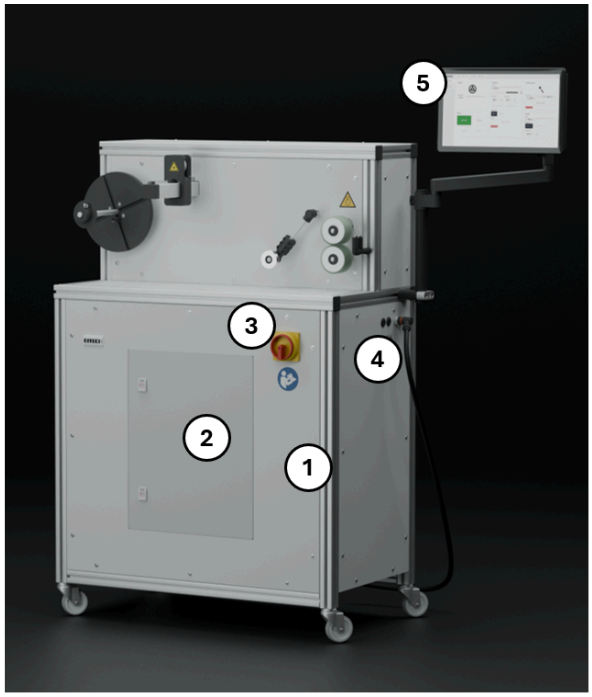
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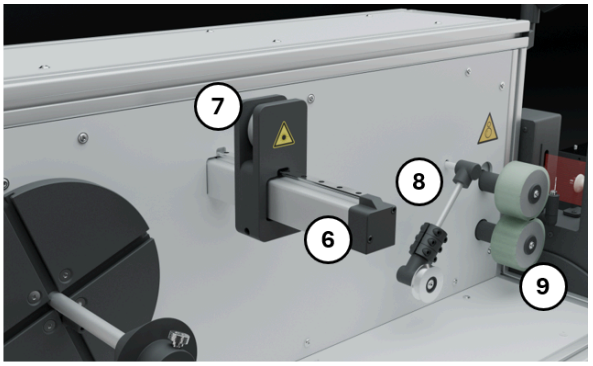
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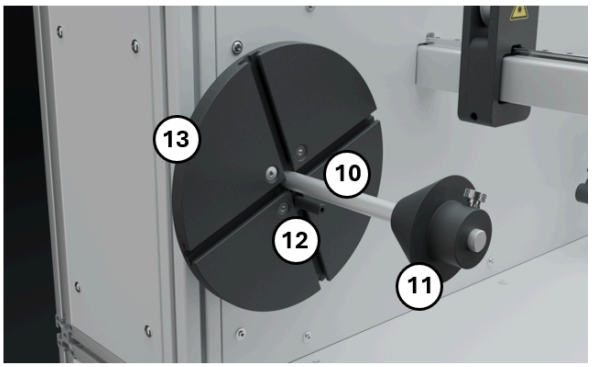
1. Machine description

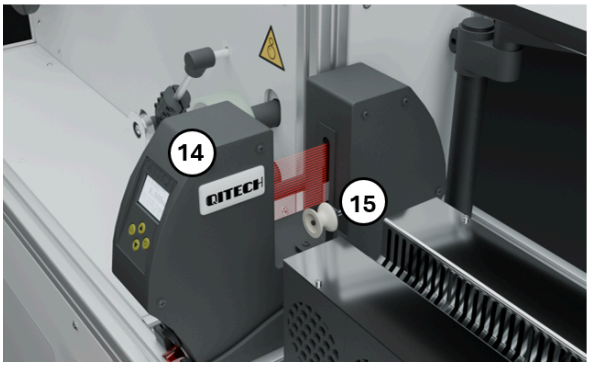
These operating instructions apply to both the QiTech Winder and the QiTech Winder XL. The two variants are identical in function and operation; the XL variant accommodates larger and heavier spools and differs in overall dimensions and weight. All specifications that differ between the two variants are indicated separately in section 1.2 Technical specifications.

1.1 Component overview













Item number	Description
1	Underframe with heavy-duty casters
2	Electrical control cabinet
3	Main switch with emergency switching-off function

4	Connector ports
5	Control display
6	Installer for traverse
7	Traverse with laser diode
8	Load arm with extra weights
9	Puller rollers
10	Spool axis
11	Spool quick-release fastener
12	Grooved plate
13	Spool fixation help
14	2-axis laser measurement device
15	Filament guide rolls
16	Power-supply-cable (5 pin M12)
17	LAN-cable

The design of the main control element depends on how the QiTech Winder is operated.

In standalone operation, the QiTech Winder is equipped with a red/yellow rotary main switch, which serves as both the main switch and the emergency switching-off device for the machine.

When the QiTech Winder is operated as part of a process line together with the QiTech Extruder and the QiTech Aquapath or QiTech Airpath, a central emergency stop is provided at the cooling unit for easy access from the front of the process line. It switches off the entire process line and needs to be released manually before the process line can be restarted. The main switch on the QiTech Winder retains the same function, but it no longer serves as an emergency switching-off device. It functions solely as the main switch of the machine and switches off the QiTech Winder only.

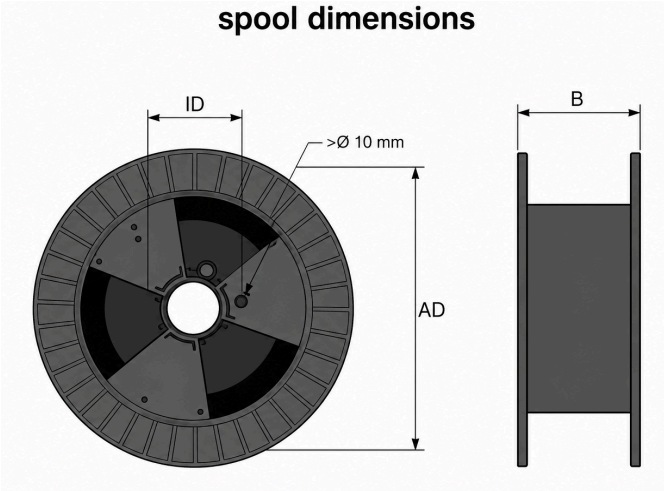
The arrangement of the machines and control elements in the process line is shown in the rendering in Annex A.

1.2 Technical specifications

Model	Winder	Winder XL
Dimensions and weight		
Length x Width x Height	1220 mm x 700 mm x 1350 mm	1000 mm x 510 mm x 1350 mm
Weight	76 kg	88 kg
Electrical specifications		
Rated voltage	230 V	
Rated frequency	50 Hz	
Rated apparent power	0,759 kVA	
Rated current	4 A	
Process parameters		
Permissible filament diameter	1 – 3 mm	
Permissible filament temperature	50 °C	
Permissible spool dimensions (min.); ID x B x AD	12 mm x 32 mm x 150 mm	12 mm x 32 mm x 150 mm
Permissible spool dimensions (max.); ID x B x AD	70 mm x 130 mm x 380 mm	70 mm x 300 mm x 430 mm
Winding speed (max.)	50 m/min	
Noise emissions		

Workplace-related emission value (L_pA)	< 50 dB

The approved dimensions of the spools are shown in the following figure for better understanding.

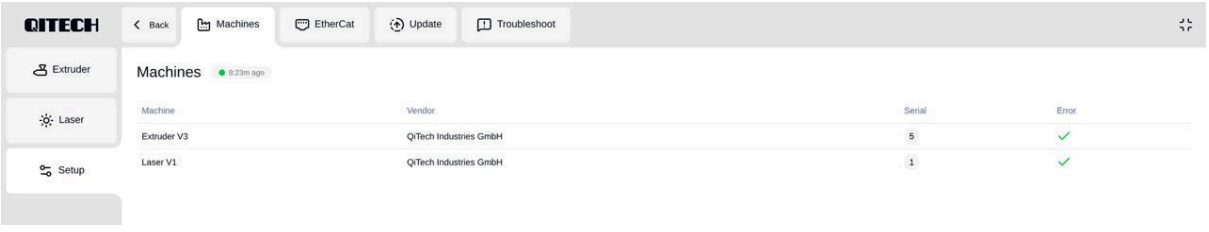


1.3 Controls and interface

1.3.1 General controls and interface

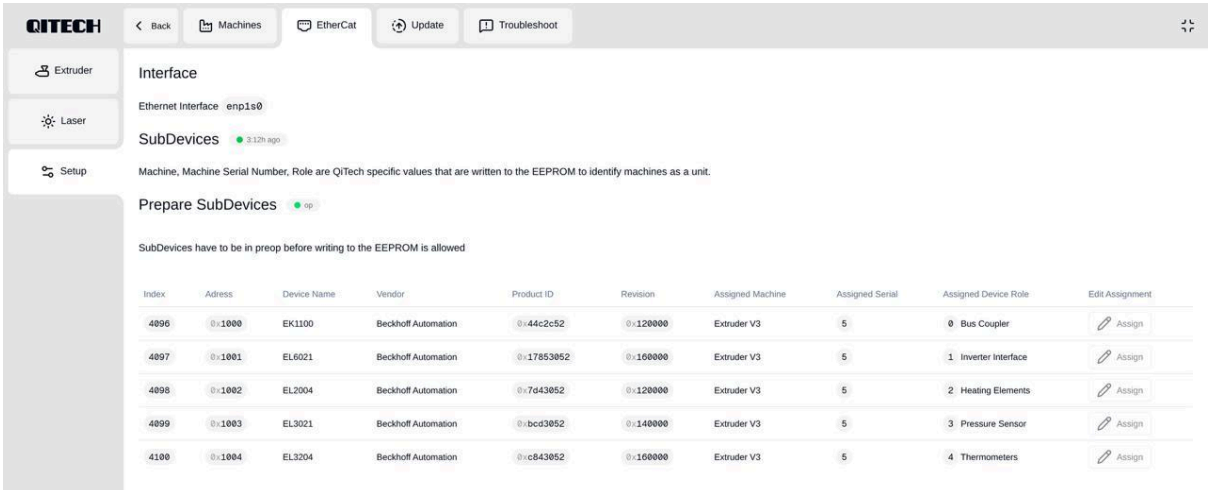
The QiTech Winder is operated entirely via the QiTech Control software. The machine is accessible on the left-hand sidebar, which makes it possible to control multiple machines simultaneously.

The sidebar on the left switches between the views of the connected machine. *Set Up* provides access to machine identification, EtherCAT device configuration, software updates and system troubleshooting functions.



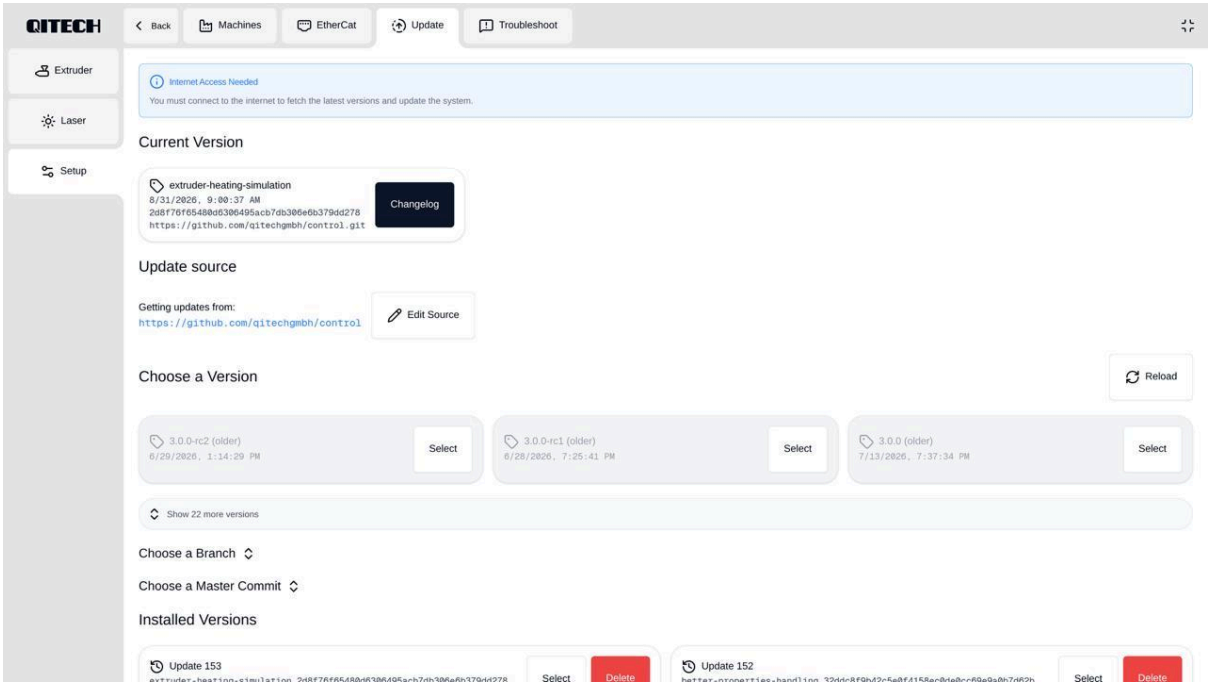
Machines	
Machine	Shows the name and version of the configured machine

Vendor	Shows the manufacturer of the respective machine
Serial	Displays the assigned machine serial number
Error Status	Indicates whether an error is currently detected for the machine



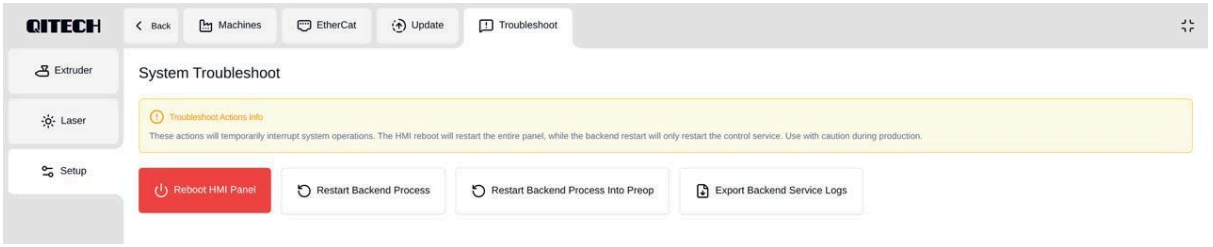
Ethercat	
Ethernet Interface	Shows the Ethernet interface currently used for EtherCAT communication
SubDevices	Displays all detected EtherCAT subdevices connected to the system
Index	Shows the internal index of the respective EtherCAT subdevice
Address	Shows the EtherCAT address assigned to the subdevice
Device Name	Displays the detected EtherCAT device type
Vendor	Shows the manufacturer of the EtherCAT device

Product ID	Displays the EtherCAT product identification number
Revision	Displays the hardware/device revision
Assigned Machine	Shows the machine to which the subdevice is assigned
Assigned Serial	Shows the serial number of the assigned machine
Assigned Device Role	Shows the function of the subdevice within the machine
Edit Assignment	Opens the assignment settings for the selected subdevice



Update	
Current Version	Displays the software version currently installed and running
Changelog	Opens the changelog of the current software version

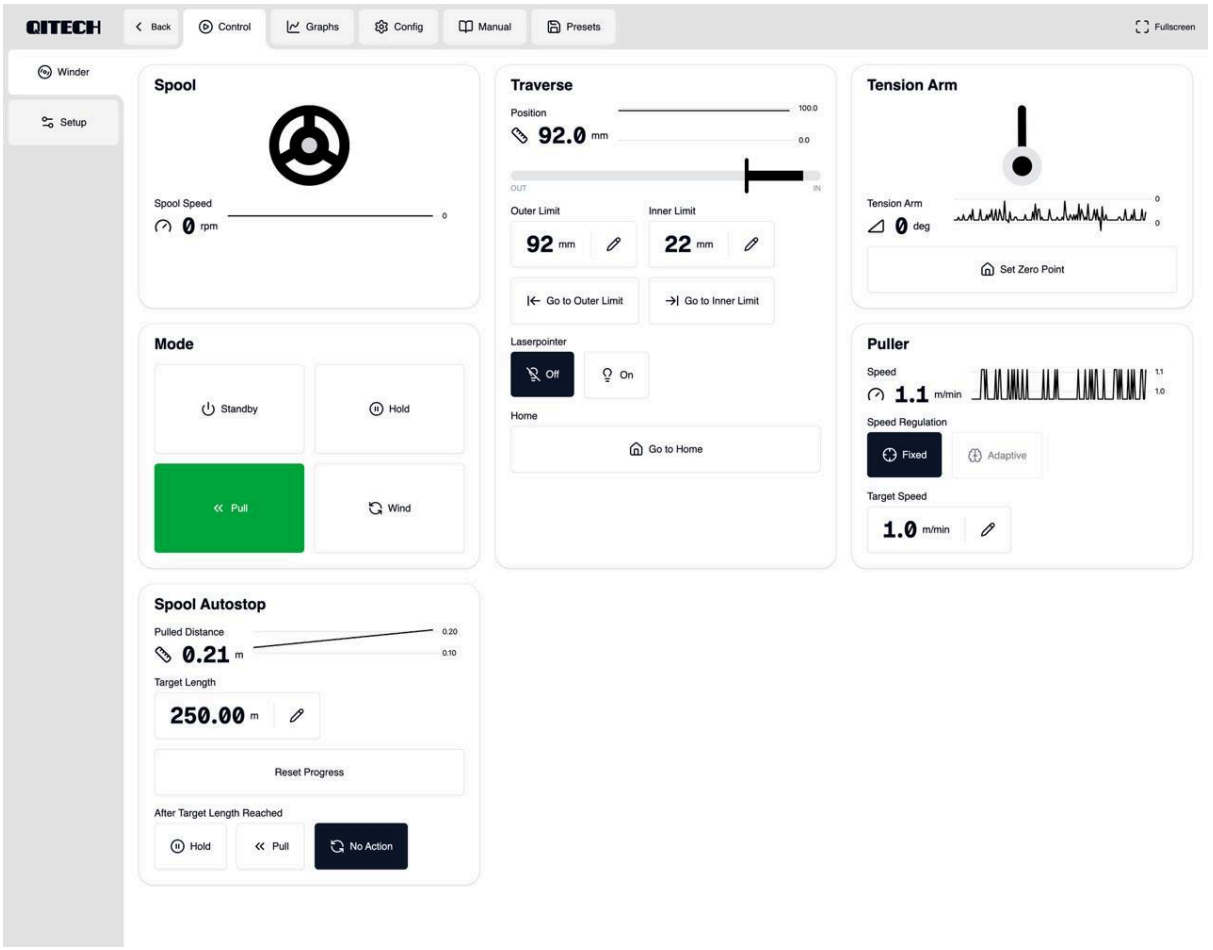
Update Source	Shows the repository or source from which software updates are retrieved; Edit source if software update source must be changed
Choose a Version	Displays available software versions that can be installed; Select the corresponding software version for installation
Reload	Refreshes the list of available software versions
Choose a Branch	Allows a software branch to be selected manually
Choose a Master Commit	Allows a specific master commit to be selected manually
Select Installed Version	Displays software versions already stored on the system; Select or delete a previously installed software version



Troubleshoot	
Reboot HMI Panel	Reboots the complete HMI panel and restarts the user interface
Restart Backend Process	Restarts the backend control process without rebooting the complete HMI panel
Restart Backend Process Into PREOP	Restarts the backend process and places the EtherCAT system into PREOP state
Export Backend Service Logs	Exports backend service log files for troubleshooting and diagnostic purposes

1.3.2 Machine-specific controls and interface

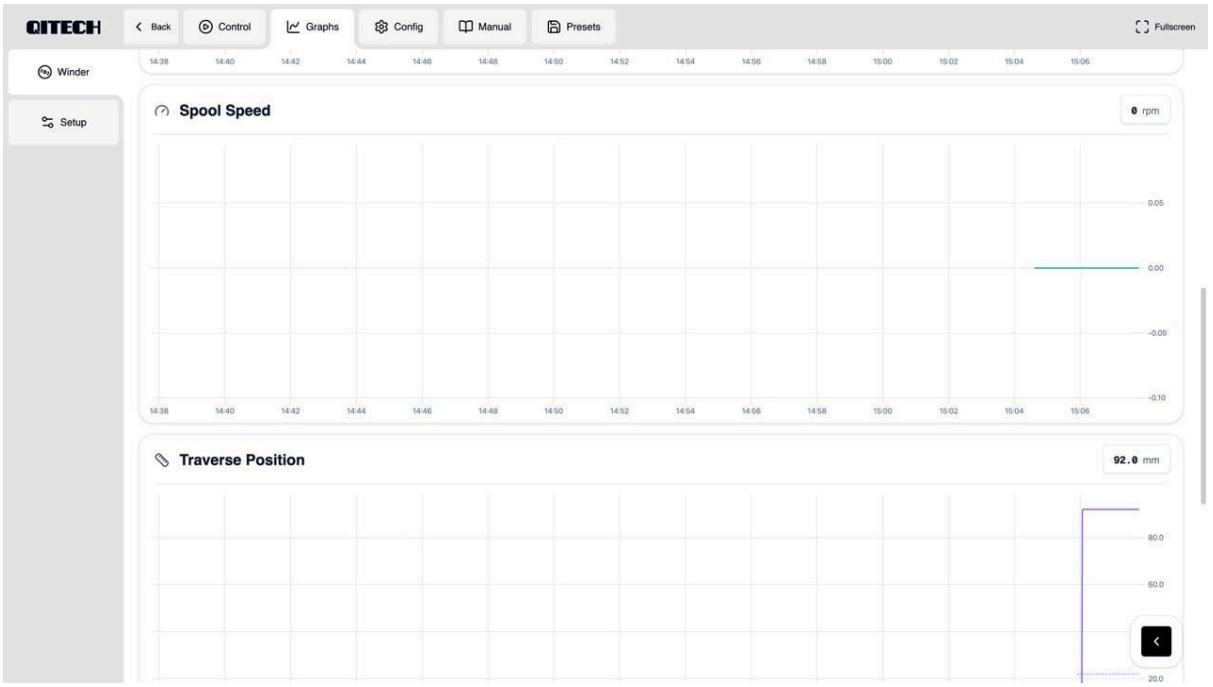
The navigation bar at the top of the screen provides access to the main areas of the QiTech Control software. The *Control* view is the primary operating view, displaying all real-time machine parameters. The *Graphs* view shows process data graphically over time and allows data to be exported in Excel format. The *Config* view provides access to the machine configuration settings. The *Manual* view contains this operating manual, so that all instructions remain available directly at the machine. The *Presets* view is used to save the current parameter set and to load previously stored ones, which allows a proven set of settings to be recalled for a known material. The *Back* button returns to the machine selection screen, and *Fullscreen* expands the software to the full display area.



Control	
Spool	
Spool speed	Displays the current speed of the spool axis and the estimated diameter; adjusted automatically by the control system

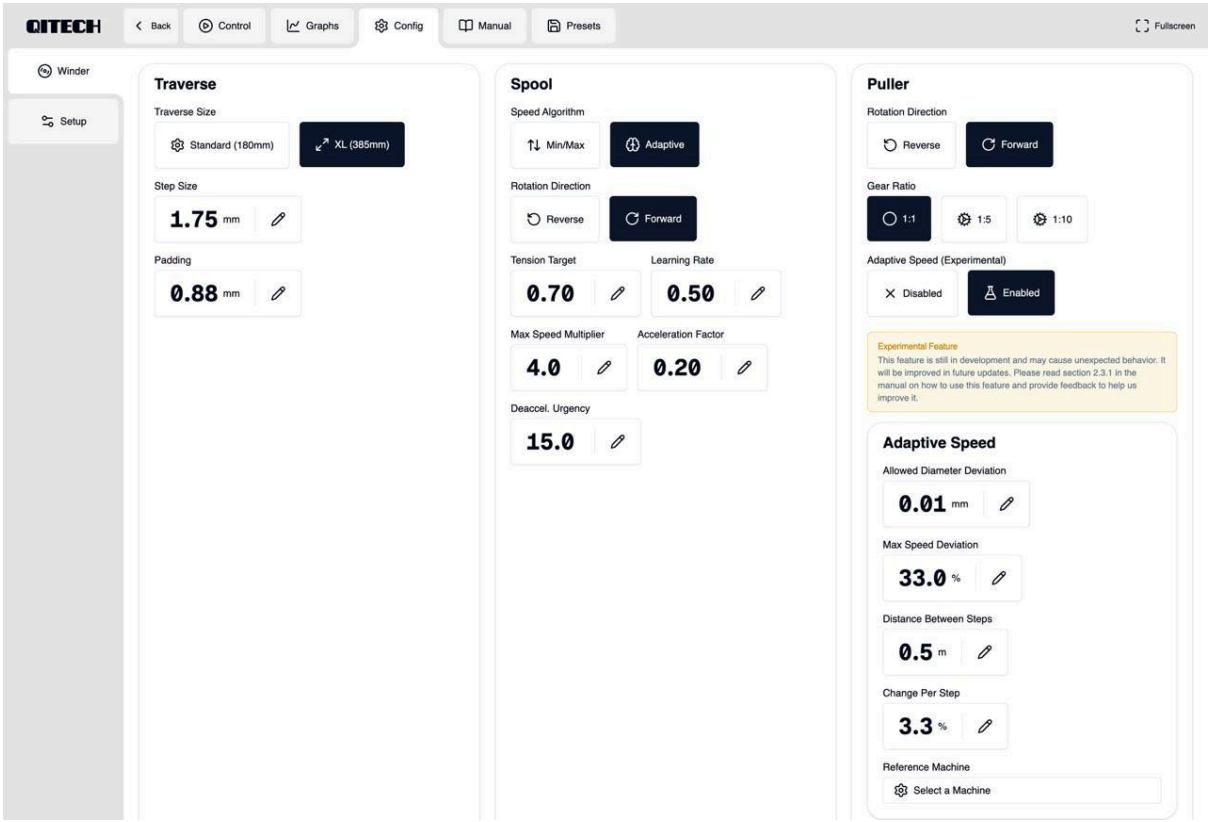
Traverse	
Position	Displays the current position of the traverse
Outer limit	Defines the outer boundary of the traverse movement, corresponding to the outer flank of the mounted spool; adjustable via the edit button
Inner limit	Defines the inner boundary of the traverse movement, corresponding to the inner flank of the mounted spool; adjustable via the edit button
Tension arm	
Tension arm	Displays the current angle of the load arm in degrees; used by the control system to regulate spool and puller speeds
Set Zero Point	Sets the current load arm position as the reference zero point for tension regulation
Mode	
Standby	Cuts all motor current; no machine movement
Pull	Activates the puller rollers; the spool axis remains stationary
Wind	Activates full winding operation; puller rollers, spool axis and traverse all operate simultaneously
Hold	Keeps all motors energised but stops all movement to pause operation temporarily
Puller	
Speed	Displays the current puller speed

Regulation	Toggles between two control modes: Speed and Diameter
Target Speed	Sets the target puller speed; adjustable via the edit button



Graph	
Puller Speed	Displays the puller roller speed in m/min over time, allowing the operator to monitor consistency of filament transport during the winding process
Tension Arm Angle	Displays the load arm angle in degrees over time, reflecting the filament tension during operation
Spool Speed	Displays the rotational speed of the spool axis in rpm over time; the spool speed adjusts automatically
Traverse Position	Displays the current position of the traverse in mm over time, showing the back-and-forth movement of the traverse unit along the spool width
Spool Progress	Displays the total length of filament wound onto the spool in metres over time

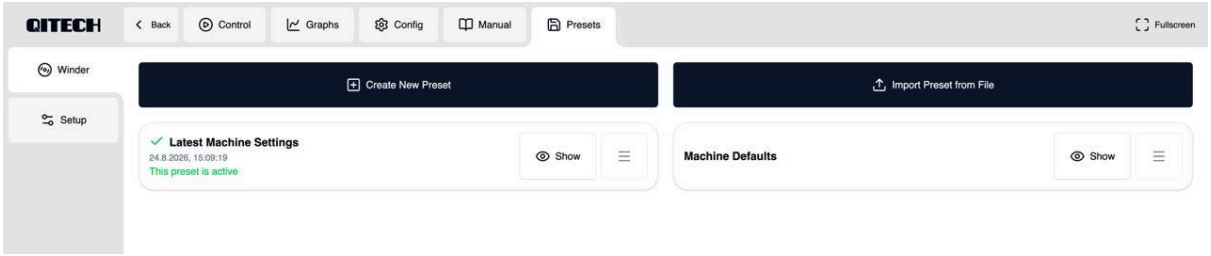
The graphs can be displayed either as a historical view or as a live view that updates continuously. A time range of up to one hour can be selected. Using the "Add Marker" and "Remove Marker" buttons, specific points in time can be marked manually, named and displayed in the graph. The "Export" button allows the process data to be exported in Excel format.



Config	
Traverse	
Traverse Size	Selects the traverse range corresponding to the installed machine variant: Standard or XL; this setting must match the physical machine variant in use.
Step Size	Defines the distance in mm between adjacent filament windings on the spool
Padding	Defines an additional margin in mm applied at each end of the traverse range to prevent the filament from being wound too close to the spool flanks
Spool	
Speed Algorithm	Selects the algorithm used to regulate spool speed. *Min/Max* switches between a

	minimum and maximum speed range. *Adaptive* continuously adjusts the spool speed based on the tension arm angle
Rotation Direction	Sets the rotation direction of the spool axis: *Forward* for winding, *Reverse* for unwinding
Tension Target	Sets the target tension arm angle used by the Adaptive speed algorithm as the reference value for tension regulation
Learning Rate	Defines the rate at which the Adaptive speed algorithm adjusts to changes in filament tension. Higher values result in faster adaptation
Max Speed Multiplier	Defines the maximum factor by which the spool speed may be increased relative to the base speed during adaptive regulation
Acceleration Factor	Defines the rate at which the spool accelerates when the control system increases the spool speed
Deaccel. Urgency	Defines the urgency with which the spool decelerates when the control system reduces the spool speed. Higher values result in faster deceleration
Puller	
Rotation Direction	Sets the rotation direction of the puller rollers: *Forward* for winding and pulling, *Reverse* for unwinding
Gear Ratio	Selects the gear ratio of the puller drive: *1:1*, *1:5* or *1:10*. The correct setting depends on the physical configuration of the puller unit
Adaptive Speed	Enables or disables the experimental Adaptive Speed function, which automatically adjusts the puller speed to maintain a consistent filament diameter
Adaptive Speed	
Allowed Diameter Deviation	Defines the maximum permitted deviation in mm from the target filament diameter before the control system adjusts the puller speed
Max Speed Deviation	Defines the maximum permitted deviation in % from the current puller speed that the Adaptive Speed function may apply per

	adjustment step
Distance between Steps	Defines the filament length in metres between successive speed adjustment steps
Change per Step	Defines the percentage by which the puller speed is adjusted at each step
Reference Machine	Selects a reference machine whose speed profile is used as the basis for the Adaptive Speed regulation



Presets	
Create New Preset	Saves the current machine configuration as a new named preset
Import Preset from File	Loads a previously exported preset from a file
Show	Displays the full configuration parameters of the selected preset
Menu (≡)	Opens additional options for the selected preset, such as renaming, exporting or deleting

1.4 Intended use

The QiTech Winder is designed for winding and rewinding of filaments and materials with similar properties, such as cords or wires, with a diameter of 1–3 mm in industrial or commercial indoor environments. The machine must be placed on a stable, level surface with all parking brakes applied.

The following conditions of use are prohibited:

- Operation in potentially explosive atmospheres
- Operation in food production environments
- Operation outdoors
- Any use beyond the scope defined in these operating instructions

Cleaning, maintenance, troubleshooting and repair are considered part of the intended use, provided they are carried out by qualified personnel in accordance with these operating instructions.

In the event of significant modifications of the machine or deviations from the intended use, a new risk assessment must be carried out and all changes must be documented accordingly.

1.5 Reasonably foreseeable misuse

The following types of use are considered reasonably foreseeable misuse and are prohibited:

Winding of unsuitable filaments:

- Filaments outside the permitted diameter range of 1–3 mm
- Filaments with a surface temperature exceeding 50 °C
- Filaments that are difficult to guide or easily tearable
- Flat, sharp-edged or pointed filaments

Improper operation:

- Operating the winder on an uneven surface or in any position other than the intended horizontal position
- Operating the winder with parking brakes not applied
- Use of spool sizes outside the permitted dimensions (see section 1.2 Technical specifications)
- Use of spare parts or components not authorised by the manufacturer
- Operating the winder unsupervised or outside of staffed working hours
- Removal or bypassing of safety guards or manipulation of sensors

1.6 Significant modification / other use

Any structural, mechanical or electrical modification to the machine that has not been expressly authorised by QiTech Industries is considered a significant modification. Significant modifications may affect the safety of the machine, render the existing risk assessment invalid and void the CE conformity of the machine. In such cases, a new conformity assessment must be carried out before the machine is put back into service.

Any use of the machine beyond the scope defined in section 1.4 is considered improper use. QiTech Industries and its authorised representatives accept no liability for damage to persons or property resulting from improper use or unauthorised modifications. In such cases, liability rests solely with the operator.

2. Safety

2.1 General safety instructions

These safety instructions apply to all persons working on or with the QiTech Winder. All personnel must be familiar with these instructions and with the information in the following chapters.

Personnel and authorisation

The machine may only be operated by professionally trained and instructed persons. Uninstructed persons and children must not have access to the machine. Repairs and maintenance work may only be carried out by qualified specialist personnel who are familiar with the relevant occupational safety regulations. Work on the electrical equipment of the machine may only be carried out by a qualified electrician.

Personal protective equipment

Appropriate personal protective equipment must be worn at all times. In particular:

- Always wear safety footwear
- Wear close-fitting clothing; use a hairnet if you have long hair
- Do not wear loose clothing, ties, open shirt sleeves or jewellery
- Activities with a risk of entrapment must be carried out without gloves; if gloves are required for a specific task, the machine must first be brought to a safe standstill
- Hearing protection must be worn if the emission sound pressure level at the workplace exceeds 80 dB(A)

Electrical safety

Before carrying out any maintenance, cleaning or repair work, the machine must be switched off and fully de-energised by disconnecting the mains plug. The following steps must be observed:

- Switch off the machine and select Standby mode
- Disconnect the main plug
- Disconnect the LAN connection
- Secure the machine against unintentional restart
- Have the machine de-energised by a qualified electrician
- Earth and short-circuit the machine
- Cover or isolate neighbouring live parts

The mains supply cable may only be connected to a socket outlet with earthing contact or a suitable connection box. Before connection, have the socket outlet or connection box checked by a qualified electrician. Never use the machine with defective, crushed or bare cables.

Mechanical safety

Do not reach into moving parts of the machine. Keep a safe distance from all rotating parts during operation. Spools may only be changed when the spool axis is at a complete standstill. Never leave the machine in operation without supervision, and remain with the machine until it has come to a full standstill before leaving the work area.

General operational safety

Before starting work, carry out a visual inspection to ensure the machine is in perfect technical condition. The machine may only be used when all protective devices are properly fitted and fully operational.

- Keep the work area clean and free of obstacles; do not place objects on the machine
- Switch off the machine when it is not in use
- Protect the machine from moisture
- Check the machine regularly for damage; only use original spare parts authorised by the manufacturer
- Ensure adequate lighting in accordance with applicable national workplace regulations

QiTech Industries accepts no liability for damage caused by failure to observe these operating instructions. The use of non-authorised spare parts or accessories will invalidate the guarantee and warranty.



Danger to life due to technically faulty machine
The machine may only be used when it is in perfect technical condition, in accordance with its intended use and in compliance with these operating instructions. All faults, especially those that could impair safety, must be rectified immediately before starting the winder.

2.2 Pictograms, symbols and warning notices

This chapter provides a comprehensive reference of all safety symbols that operators may encounter in an industrial working environment. Graphical symbols, including warning signs, mandatory signs, prohibition signs, follow ISO 7010. All people working with the QiTech Winder must be familiar with the meanings of these symbols.

The general warning symbol can be used in combination with one of the following signal words:

DANGER Indicates an imminently hazardous situation which, if not avoided, will result in death or serious injury.

WARNING Indicates a potentially hazardous situation which, if not avoided, could result in death or serious injury.

CAUTION Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury.

NOTICE Indicates important information not related to personal injury. Non-observance may result in damage to property or the environment.

Warning signs		**Mandatory signs**	
	W001 General warning sign		M001 General warning sign
	W004 Warning of laser beam		M003 Use hearing protection
	W011 Warning of slippery surface		M004 Use eye protection
	W012 Warning of electrical voltage		M008 Use foot protection
	W016 Warning of toxic substances		M009 Use hand protection
	W017 Warning of hot surface		M010 Use protective clothing

	W018 Warning against automatic start-up		M012 Use handrail
	W019 Warning of crushing hazard		M014 Use head protection
	W024 Warning against hand injuries		M021 Switch off before maintenance or repair
	W025 Warning against counter-rotating rollers		
Prohibition signs		**Other pictograms**	
	P001 General prohibition sign		F001 Fire extinguisher
	P002 No smoking		WEEE symbol Separate collection of waste electrical and electronic equipment
	P003 No open flames; fire, open source of ignition and smoking not permitted		

	P022 Eating and drinking not permitted		
	P024 Entering the area not permitted		
	P031 Switching not permitted		

2.3 Safety devices

The QiTech Winder is equipped with the following safety devices. All safety devices must be fully operational before the machine is started. It is not permitted to operate the machine with any safety device removed, bypassed or disabled. If a safety device is opened or removed for maintenance purposes, it must be correctly refitted and verified before the machine is switched on again.

Main switch with emergency switching-off function
The machine is switched on and off with the rotary main switch on the front of the machine. In addition to its function as the main switch, it serves as an emergency switching-off device, indicated by the red and yellow colour. In the event of danger it can be turned to the OFF position at any time, disconnecting the machine from the main supply. It is accessible from all operating positions in front of the machine. When switched off, the drives are de-energised and all hazardous movements stop (uncontrolled stop, stop category 0 as per IEC 60204-1). The machine can only be restarted by turning the switch back to the ON position and subsequently selecting an operating mode in the QiTech Control software.

Spool stop in the event of filament breakage
If the filament tears during winding, the spool axis stops automatically. This prevents uncontrolled unwinding of material and reduces the risk of entanglement.

Machine housing
The machine housing, clad with aluminium composite panels, serves as a separating protective device. It prevents direct contact with hot motor components and the drive spindle of the traverse, restricts access to the electronics, and prevents electrostatic charges from transferring from the filament to the electronic components.

Notice: If safety-relevant changes or defects are detected on the machine, shut it down immediately, secure it against restart and report the issue to the responsible person before resuming operation.

2.4 Qualification of personnel / instruction and training

All persons working on or with the QiTech Winder must have the qualifications appropriate to their assigned tasks. The following personnel categories are defined for the purposes of these operating instructions:

Operator: The operator is the person who uses, adjusts and monitors the machine in day-to-day operation. The operator is characterised by practical experience in handling the machine as well as in-depth knowledge of its functions and the correct course of action in the event of a fault.

Instructed person: An instructed person has been briefed by the operator on the tasks assigned to them and on the potential hazards associated with improper behaviour. The correct execution of all activities carried out by an instructed person is monitored by specialised personnel.

Specialised personnel: Specialised personnel are able to carry out their assigned tasks independently and to recognise and avoid potential hazards, based on their specialist training, knowledge and experience as well as their familiarity with the relevant regulations. It is assumed that specialised personnel have also received the instruction described above.

Qualified electrician: A qualified electrician is a person who, by virtue of their technical training, knowledge and experience as well as their familiarity with the relevant regulations, is able to assess the work assigned to them and recognise and avoid potential hazards related to electrical systems.

Tasks	**Minimum qualification**
Transport	Specialised personnel of the manufacturer
Installation and commissioning	Specialised personnel of the manufacturer
Day-to-day operation	Instructed personnel of the operator
Setting, adjusting and restarting	Specialised personnel of the operator or manufacturer
Troubleshooting and fault rectification	Specialised personnel of the operator or manufacturer

Cleaning and maintenance	Specialised personnel of the operator or manufacturer
Decommissioning	Specialised personnel of the operator

As the operator, you are obliged to ensure that all personnel working on or with the machine are trained and instructed in accordance with these operating instructions before starting work. This includes instruction on all applicable legal and accident prevention regulations as well as on all safety devices on and around the machine. The varying technical qualifications of individual employees must be taken into account when planning instruction sessions.

Instruction must be understood and retained by all personnel. Compliance with instruction content must be verified regularly. Written confirmation of attendance must be obtained from each employee following each instruction session.

Restrictions on employment

The following persons must not work on or with the machine under any circumstances:

- Persons with skin diseases
- Persons with chronic respiratory conditions or a tendency to allergic reactions
- Pregnant women and nursing mothers
- Persons under 18 years of age

2.5 Residual risks

Despite the safety measures integrated into the design and construction of the QiTech Winder, certain residual risks cannot be fully eliminated. The following hazards remain and must be taken into account by all persons working on or with the machine.



Danger! Risk of entanglement and pull-in by moving parts
Loose hair, clothing or jewellery can be caught and pulled in by moving machine parts at any time during operation. Always wear close-fitting clothing and tie back long hair. Remove jewellery before starting work.



Danger! Operation only permitted with all protective devices fitted
The machine must only be operated with all protective devices correctly installed and fully functional. The operator is responsible for ensuring this before each use.



Danger! Risk of injury from tensioned filament
The filament strand between the puller rollers and the spool is under tension during operation and must not be touched. Do not process pointed, sharp-edged or particularly thin materials.



Danger! Electric shock
Before carrying out any repair, set-up or maintenance work, the machine must be fully de-energised by disconnecting the mains plug. Do not resume work until the absence of voltage has been confirmed.



Danger! Electrostatic charge on puller rollers and spool
Friction of the filament on the puller rollers and spool can generate electrostatic charges, which may result in minor electric shocks. Do not touch the affected areas during operation; maintain a safe distance.



Danger! Automatic speed adjustment of puller, spool and traverse
The spool speed, puller speed and traverse speed adjust automatically during operation to maintain constant winding tension. Unexpected speed changes may occur at any time. Never reach into the machine during operation.



Danger!
Risk of body parts being drawn in at the puller rollers and spool axis
The areas around the puller rollers and spool axis present an risk of entanglement and pull-in. Always wear close-fitting clothing, keep long hair tied up and do not wear long jewellery. In case of emergency, turn the main switch with emergency switching-off function immediately.



Danger! Risk of crushing when threading the filament
Exercise extreme caution when threading the filament through the puller rollers. There is a risk of crushing between the upper and lower pair of rollers.



Danger! Risk of crushing between traversing unit and machine housing
Exercise extreme caution when the traversing unit is in motion. Do not place any limbs in the gap between the traversing unit and the machine housing.

3. Transport, installation and commissioning

3.1 Transport

Warning! Risk of injury due to improper transport
When loading and unloading the machine, do not place any limbs under machine parts when setting them down. Do not stand under suspended loads or between moving loads and walls or other machine parts. Wear appropriate personal protective equipment during all transport operations: safety footwear, protective work clothing and protective gloves.

The QiTech Winder is delivered fully assembled on its mobile underframe.

Before transport, the following steps must be carried out:

- Switch off the machine and disconnect the mains plug
- Secure all loose cables and accessories to prevent damage during transport
- Open all parking brakes on the underframe before moving the machine

The machine may be moved manually on its castors over short distances on a level surface. For longer distances or loading onto a vehicle, suitable lifting equipment must be used. The permissible load-bearing capacity of the lifting equipment must meet the machine weight. After transport, inspect the machine for any damage that may have occurred during transit before putting it into service.

Caution! Risk of damage due to incorrect installation and commissioning
Errors during installation and commissioning can cause considerable damage to property. Installation and commissioning may only be carried out by the manufacturer's employees.

3.2 Requirements for the installation site

Before installing the machine, ensure that the installation site meets the following requirements:

- The surface must be level, solid and capable of supporting the weight of the machine
- The location must be used exclusively for industrial or commercial purposes
- The location must be as fire-free as possible and equipped with a fire extinguisher
- Sufficient clearance must be maintained around the machine to allow safe operation and unobstructed access for maintenance work
- Adequate lighting must be provided in accordance with applicable national workplace regulations

3.3 Basic installation

The QiTech Winder is delivered fully assembled on its mobile underframe. After positioning the machine at the intended installation site, carry out the following steps:

- Position the machine at the intended installation site in accordance with the requirements
- Apply all parking brakes to secure the machine against unintentional movement
- Route all cables in a way that they cannot be tripped over
- Ensure that the main switch with emergency switching-off function is freely accessible from all positions in front of the machine

The machine is delivered with the latest available version of the QiTech Control software installed.

3.4 Electrical connection

Danger! Electric shock due to improper electrical connection
All work on the electrical connection must be carried out by a qualified electrician. Ensure the mains socket is de-energised before connecting the machine.

Before connecting the machine to the mains supply, carry out the following steps:

- Verify that the mains voltage and frequency at the installation site correspond to the specification of the machine
- Ensure that the mains socket is equipped with an earthing contact and has been checked by a qualified electrician
- Only use original fuses with the prescribed current ratings to avoid overloading
- Connect the mains plug to the earthing contact socket

Ensure that proper earthing of the electrical system is maintained through protective conductor systems at all times.

3.5 Commissioning

Before switching on the machine for the first time, carry out a full visual inspection to verify that:

- All protective devices are correctly fitted
- All cables are undamaged and correctly routed
- The main switch with emergency switching-off function is in its released position
- The parking brakes are applied

Once all checks have been completed satisfactorily, the machine may be switched on.

4. Operation

4.1 Starting the machine

A visual inspection must be carried out every working day before the machine is started. Verify that all protective devices are correctly fitted, all cables are undamaged and the main switch with emergency switching-off function is in its OFF position.

- Connect the 220 V power supply
- Switch the main switch to the ON position
- Turn on the display by pressing the button on the right-hand side of the display unit, above the USB connectors
- The Winder will appear in the left column of the QiTech Control software

4.2 Mounting, calibrating and changing the spool

- Remove the quick-release fastener from the spool axis
- Move the installer in the groove to the height of the hole in the spool flank
- Slide the spool over the axis and the installer
- Tighten the quick-release fastener firmly

Warning! Risk of injury from rotating spool axis
The spool may only be mounted or removed when the spool axis is at a complete standstill. Never reach into the rotating spool axis area during operation.

Calibration must be carried out each time a new spool is mounted, as spool dimensions may vary.

- Select *Control* in the function bar of the QiTech Control software
- Switch to *Hold* mode
- Move the *Outer Limit* slider to the approximate position of the outer spool flank using the laser dot projected by the laser diode; fine-tune the position until the dot is exactly on the edge of the inner spool flank
- Click *Save*
- Repeat the same process for the other outer spool flank

To change the spool during operation, press the *Pull* button to pause the winding process, then remove the current spool and insert the new one. Before resuming, calibrate the spool sides of the new spool. Once calibration is complete, press the *Wind* button to continue the winding process.

4.3 Winding

The QiTech Winder supports three operating modes: winding, unwinding and rewinding. Winding transfers filament from a production line or supply source onto a spool. Unwinding feeds filament from a spool back through the puller rollers. Rewinding transfers filament from a full spool onto an empty spool.

Winding

To start the winding process, proceed as follows:

- Select *Control* in the function bar
- Press the *Pull* button
- Carefully thread the filament through the puller rollers, taking care to avoid the counter-rotating rollers
- Use the *Puller Target Speed* slider to adjust the speed until the filament reaches the desired diameter
- Rotate the load arm upwards and place it on the puller
- Press the *Wind* button
- Thread the filament coming out of the puller through the filament guide and fix it to the spool
- Place the load arm on the filament strand from above

Danger! Risk of crushing when threading the filament
Exercise extreme caution when threading the filament through the puller rollers. There is a risk of crushing between the upper and lower pair of rollers. Do not wear gloves during this operation.

Unwinding

- Mount the filament spool to be unwound onto the spool axis
- Select *Config* in the function bar
- Press the *Reverse* button to switch to reverse mode
- Return to *Control* in the function bar and adjust the *Puller Target Speed* as required
- Feed the filament through the puller rollers in the direction opposite to winding, taking care to avoid the counter-rotating rollers

Rewinding

- Mount the full spool in front of the puller rollers so that it can spin freely
- Mount the empty spool on the spool axis
- Calibrate the traverse limits
- Guide the filament from the full spool over the filament guide and fix it to the empty spool
- Select *Control* in the function bar
- Press the *Wind* button
- Adjust the pulling speed using the slider

All process variables can be adjusted during the winding process without interrupting it. To apply a change, modify the value in the relevant field in the QiTech Control software and press the "Wind" button again to confirm.

4.4 Stopping and switching off

To stop the winding process temporarily, press the "Hold" button. Motors remain energised but all movement stops.

To shut down the machine completely:

- Select *Standby* mode to cut all motor current
- Wait until all moving parts have come to a complete standstill
- Switch the main switch to the OFF position
- Disconnect the mains plug

Danger! Risk of injury if the machine is stopped abruptly
If the filament becomes caught, the spool stops spinning or there is an immediate danger to persons, switch the main switch with emergency switching-off function immeadiately. Do not attempt to stop moving parts by hand.

Notice: If safety-relevant changes or defects are detected during operation, shut down the machine immediately using the main switch with emergency switching-off function, secure it against restart and report the issue to the responsible person before resuming operation.

5. Maintenance and inspection

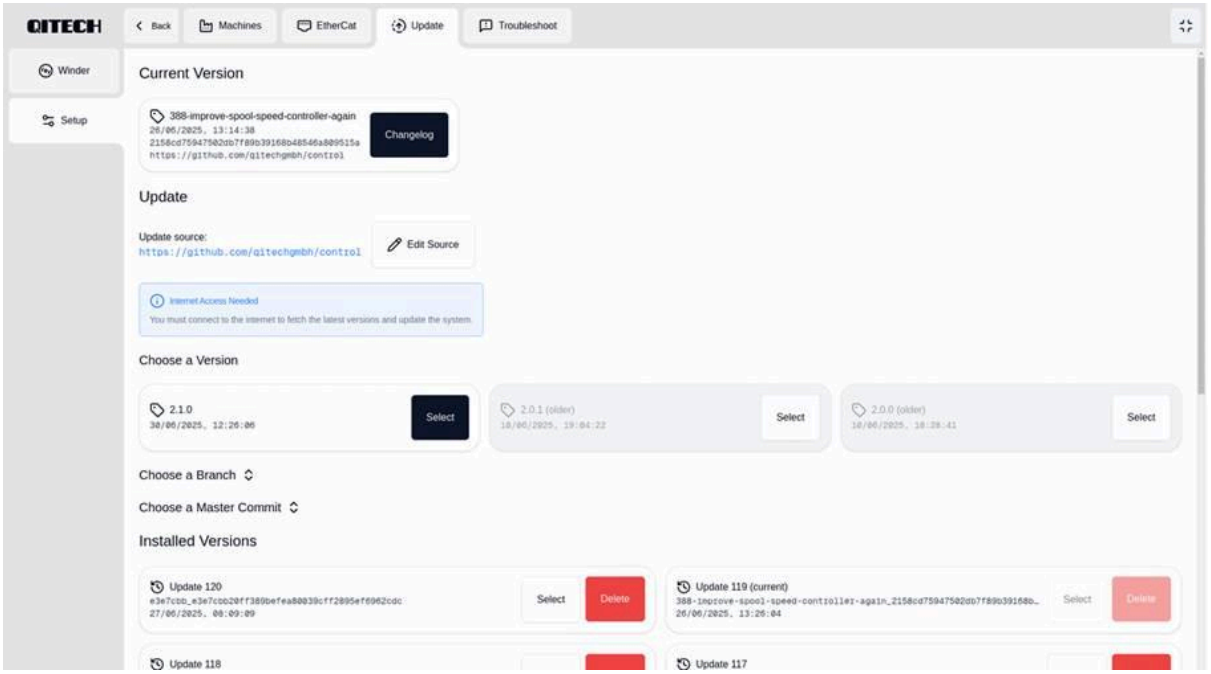
5.1 Updating the QiTech Control software

Software updates may contain bug fixes, new functions or security improvements. It is therefore recommended to check regularly whether updates are available and to install them promptly.

To perform a software update, proceed as follows:

- Ensure the machine has an active internet connection
- Open the QiTech Control software and navigate to *Setup* → *Updates*
- Available updates will be displayed; select the desired update with the *Select* button
- The display will shut down and restart automatically; the updated software version will be installed upon restart

Previous software versions and the complete update history can also be viewed under *Setup* → *Updates*.



Notice: Modifying the software source or installing unofficial software versions may compromise the safety and functionality of the machine and will invalidate the guarantee and warranty. Only official updates from QiTech Automatics GmbH may be installed.

5.2 Mechanical maintenance

Lubricating the traverse spindle

The threaded spindle of the traverse must be re-lubricated after approximately every 20 operating hours, or earlier if the traverser no longer moves reliably to the desired position while the motor is running. Proceed as follows:

- Switch off the machine and disconnect the mains plug
- Loosen the panel on top of the housing using an M5 Allen key and remove it
- Apply a lubricating oil or grease (e.g. WD40) evenly along the entire length of the threaded spindle

Retightening screw connections

After approximately every 100 operating hours, the rear panel of the machine should be removed so that the internal screw connections can be inspected and retightened. Proceed as follows:

- Switch off the machine and disconnect the mains plug
- Remove the rear panel using an M5 Allen key
- Check and retighten the following connections to at least hand-tight:
  - Grub screws of the motor couplings (puller, traverse, spool)
  - Fastening screws of the motors (puller, traverse, spool)
  - Fastening screws of the angle sensor of the load arm
- Refit the rear panel and tighten the screws securely

Replacing the puller rollers

The polyurethane coating of the puller rollers will wear out over time. The rollers should be replaced as soon as the filament begins to slip through the rollers rather than being conveyed reliably. Suitable replacement parts are available from QiTech Industries GmbH or on the commercial market. To replace the puller rollers, proceed as follows:

- Switch off the machine and disconnect the mains plug
- Unfasten the screws on the puller roller unit
- Remove the black plastic caps
- Pull out the worn puller rollers from the axes
- Slide the new puller rollers onto the axes
- Refit the plastic caps
- Refit and tighten the screws

6. Troubleshooting

The following table provides an overview of the most common faults, their possible causes and recommended corrective actions. If a fault cannot be resolved using the measures described below, contact QiTech Industries GmbH for technical support.

Before attempting to rectify any fault, ensure that the machine is in a safe condition. For faults requiring access to internal components, switch off the machine and disconnect the main plug.

Fault	Possible cause	Corrective action
Machine is not recognised by Control software	LAN cable not connected	Check and reconnect the LAN cable
	Outdated software version	Update the QiTech Control software
	Network configuration error	Check network settings
Filament tears off frequently	Filament insufficiently cooled before winding	Allow the filament to cool sufficiently before it reaches the puller rollers
	Filament outside permitted diameter range	Verify that the filament diameter is within the permitted range of 1–3 mm
	Puller speed too high	Reduce the puller target speed using the slider in the QiTech Control software
	Filament tension too high	Reduce weight on the load arm to reduce tension
Significant fluctuations of the tension arm angle	Irregular filament feed	Verify that the filament is within the permitted diameter range of 1–3 mm and that it feeds evenly. Check that the load arm moves freely and is not obstructed

	Imbalanced spool	Remove the spool and check that it is correctly seated on the spool axis and that the quick-release fastener is firmly tightened. Replace the spool if visibly damaged or deformed
Puller rollers are scored	Filament surface temperature too high when entering rollers	Ensure that the filament surface temperature does not exceed 50 °C at the puller rollers
	Filament slipping through rollers instead of being conveyed	Check whether the puller rollers require replacement
	Filament tension too high	Reduce weight on the load arm to reduce tension
Filament is not wound perfectly parallel	Inner spool diameter too large relative to traverse range	The closer the winding is to the traverse, the more uniform the winding pattern will be
	Spool sides not correctly calibrated	Recalibrate the spool sides
	Uneven first layer due to incorrect threading	Re-thread the filament carefully; an uneven first layer will cause misalignment in all subsequent layers
Winder switches off suddenly	Motor driver overheated	Wait approximately 5 minutes for the machine to cool before restarting; if the fault recurs frequently, ensure adequate ventilation at the installation site

Machine does not start	Main switch with emergency switching-off function is in the OFF position	Switch the main switch with emergency switching-off function to the ON position
	Mains plug not connected	Check the mains connection and plug in the power supply

7. Decommissioning, disassembly and disposal

When the machine has reached the end of its planned service life, QiTech Industries GmbH recommends contacting the manufacturer to arrange for the machine to be taken back. If this is not possible, the machine must be permanently taken out of service and disposed in accordance with the instructions in this chapter and in compliance with applicable national regulations.

Disassembly and decommissioning may only be carried out by trained and qualified specialist personnel of the operator.

Decommissioning

Before disassembly begins, the machine must be taken out of service by properly switching off, disconnect the power supply and secure the machine against unintentional restart. Allow the machine to cool completely before proceeding with disassembly.

Disassembly

The following tools are required for disassembly:

- M5 Allen key
- Standard screwdriver set

The following personal protective equipment must be worn during disassembly:

- Safety footwear
- Protective gloves
- Protective clothing

The machine consists largely of aluminium and plastic components, which can be easily separated by loosening the screw connections. Proceed as follows:

- Remove all cables and set aside for separate disposal
- Remove the aluminium composite panels from the housing frame by loosening the screw connections
- Separate the mechanical components from the electronic components
- Remove and separately collect all auxiliary materials, including lubricants and rubber seals

Disposal

All components must be separated by material class to enable subsequent recycling or differentiated scrapping. The following materials require separate disposal by an authorised specialist company:

- Electrical and electronic components
- Lubricants and oils
- Rubber seals

Equipment bearing the WEEE symbol must not be disposed of as unsorted waste. The operator is responsible for ensuring that all waste is separated by material and disposed of via an authorised collection or take-back point in accordance with applicable national regulations.

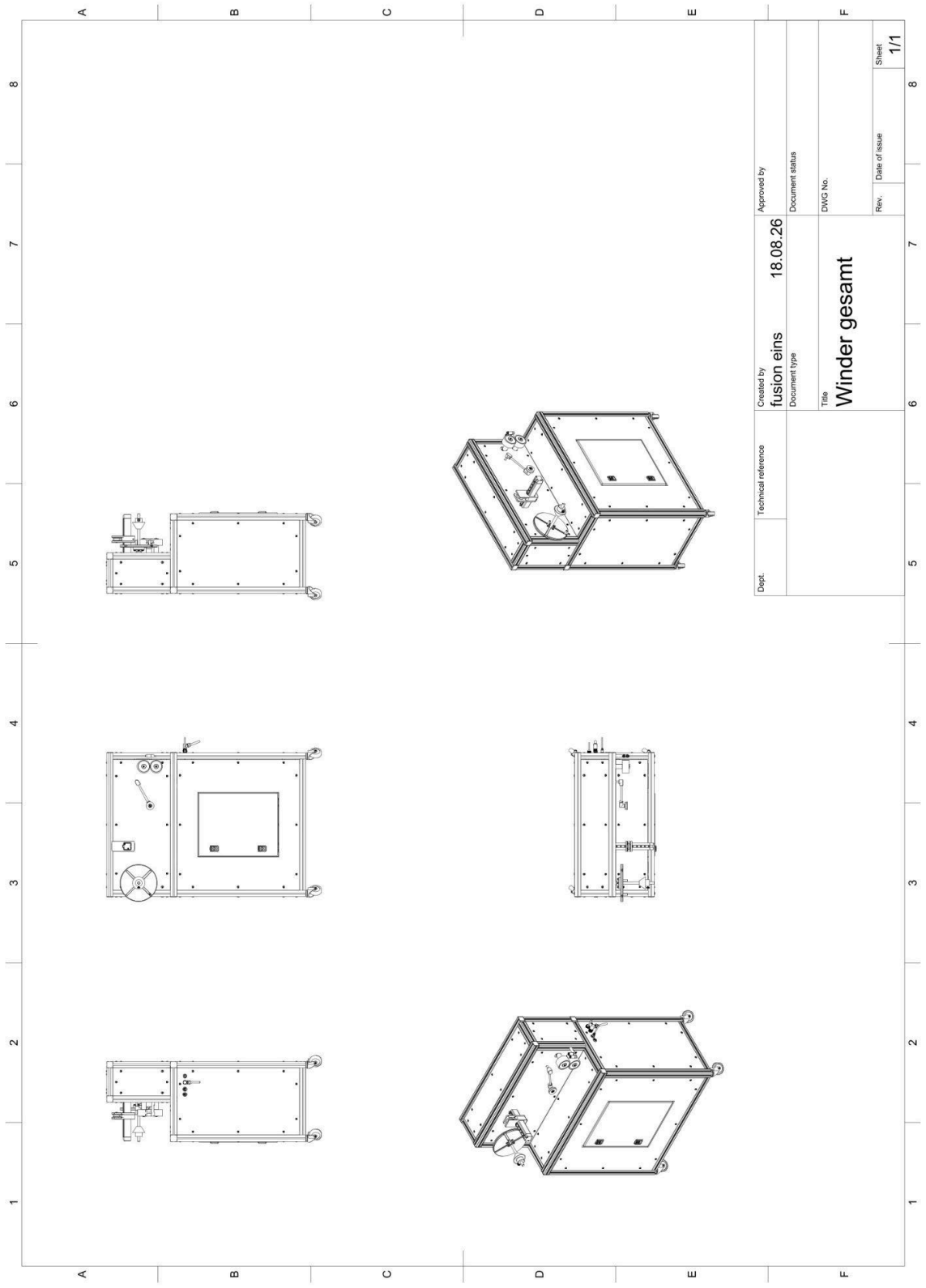
Notice: Disposal of hazardous materials such as lubricants, oils and electronic components must be carried out in accordance with applicable national environmental regulations. Contact an authorised waste disposal specialist if in doubt.

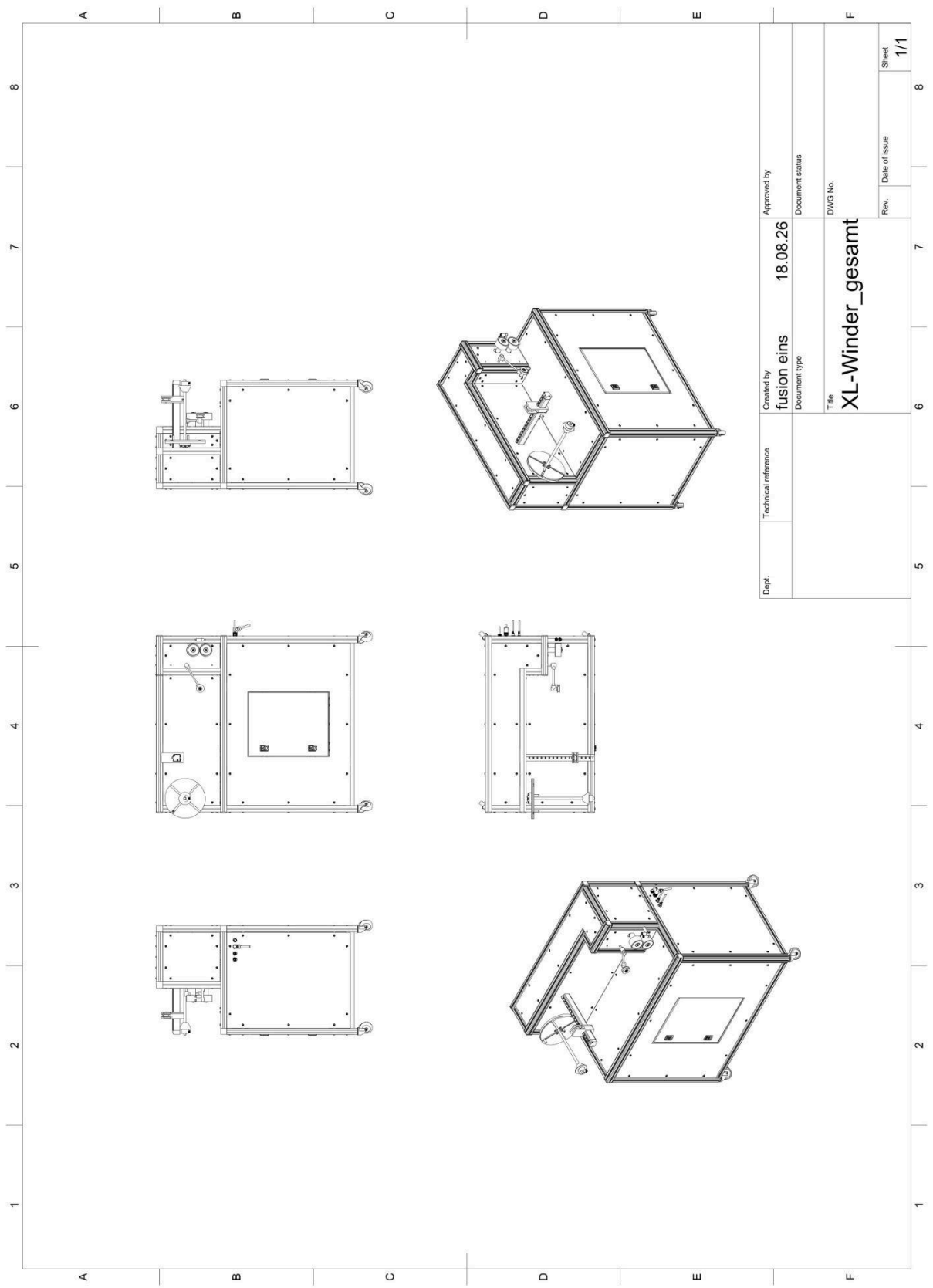
8. Annexes

Annex A: Rendering of the process line

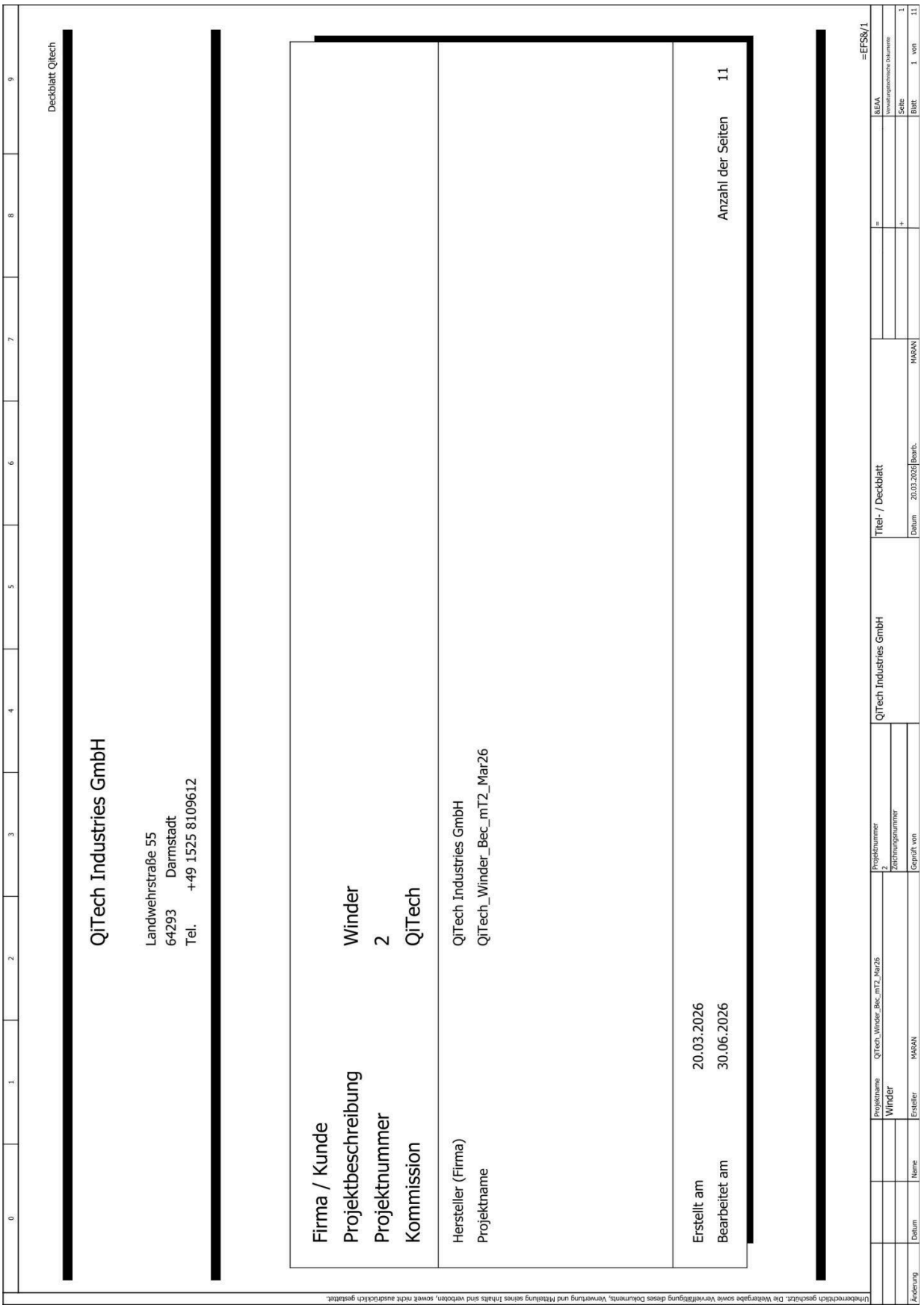


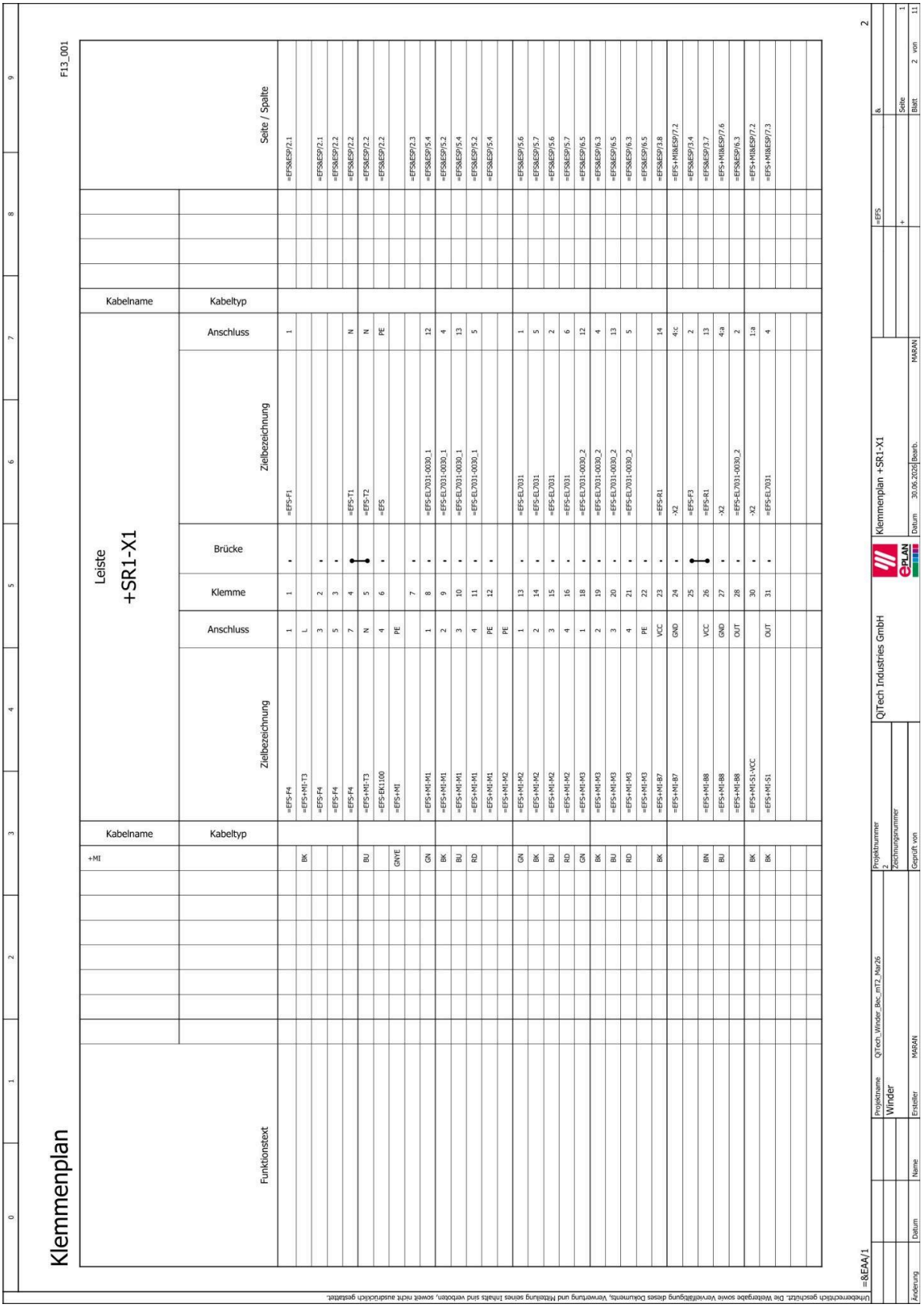
Annex B: General assembly drawing

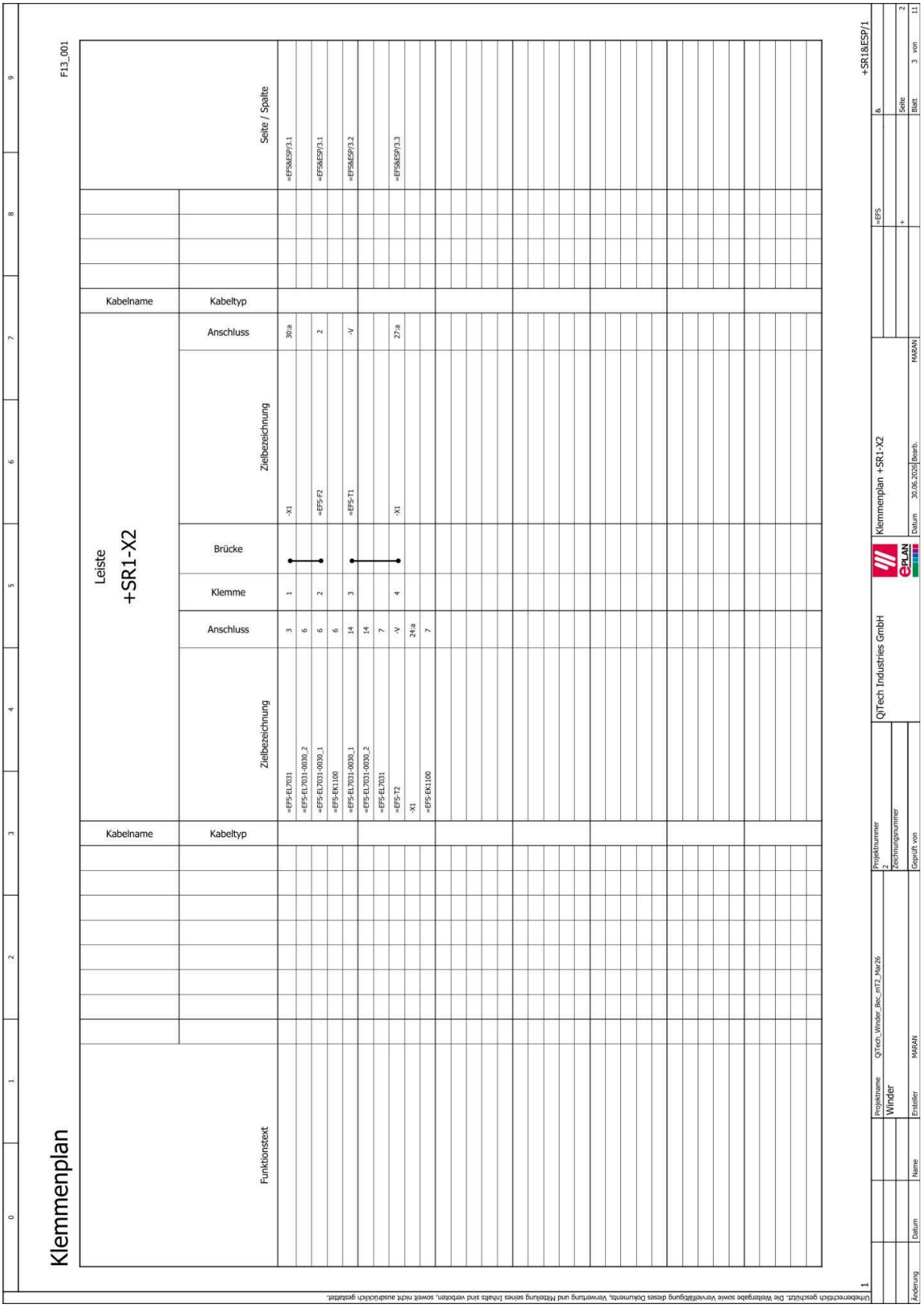


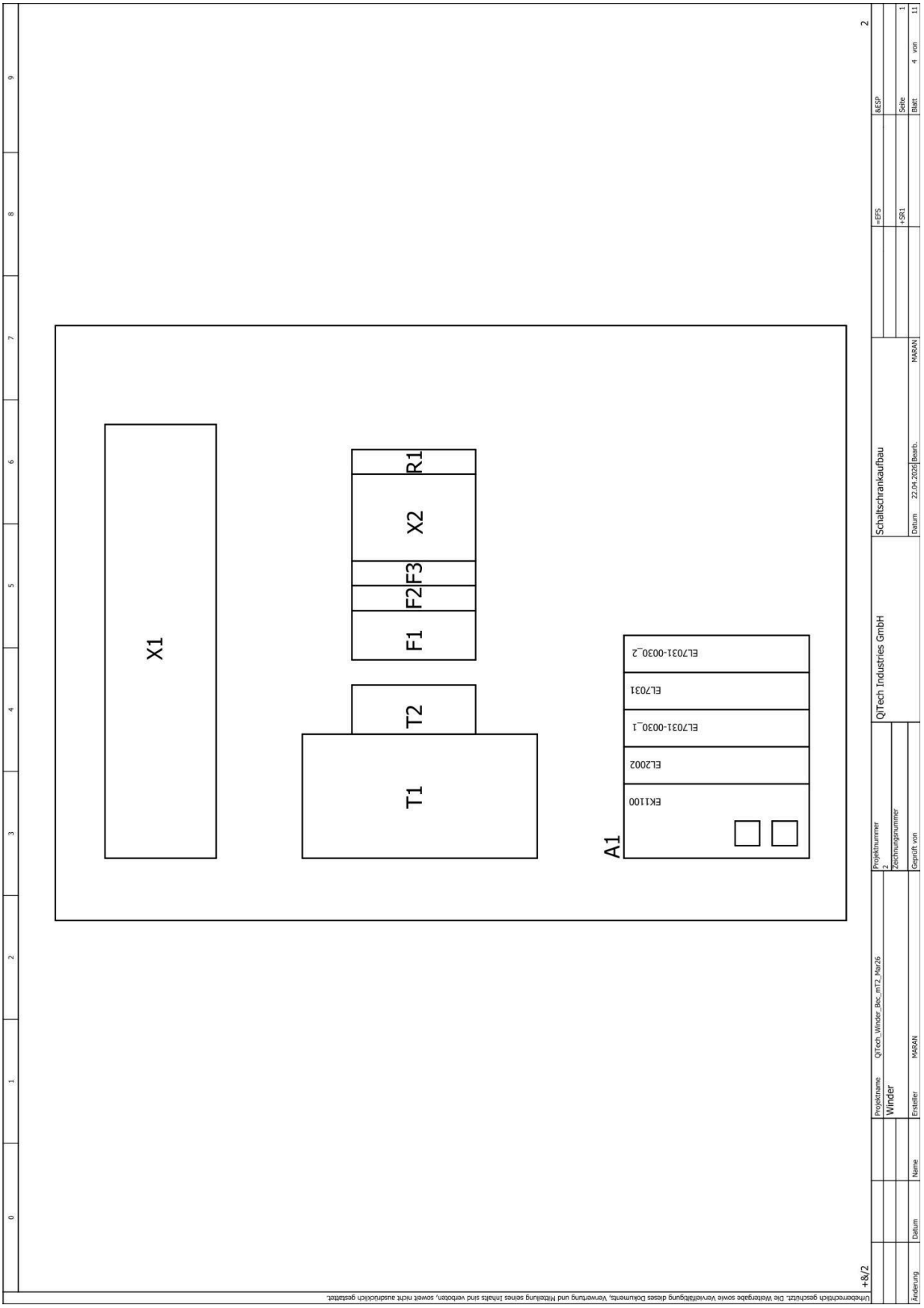

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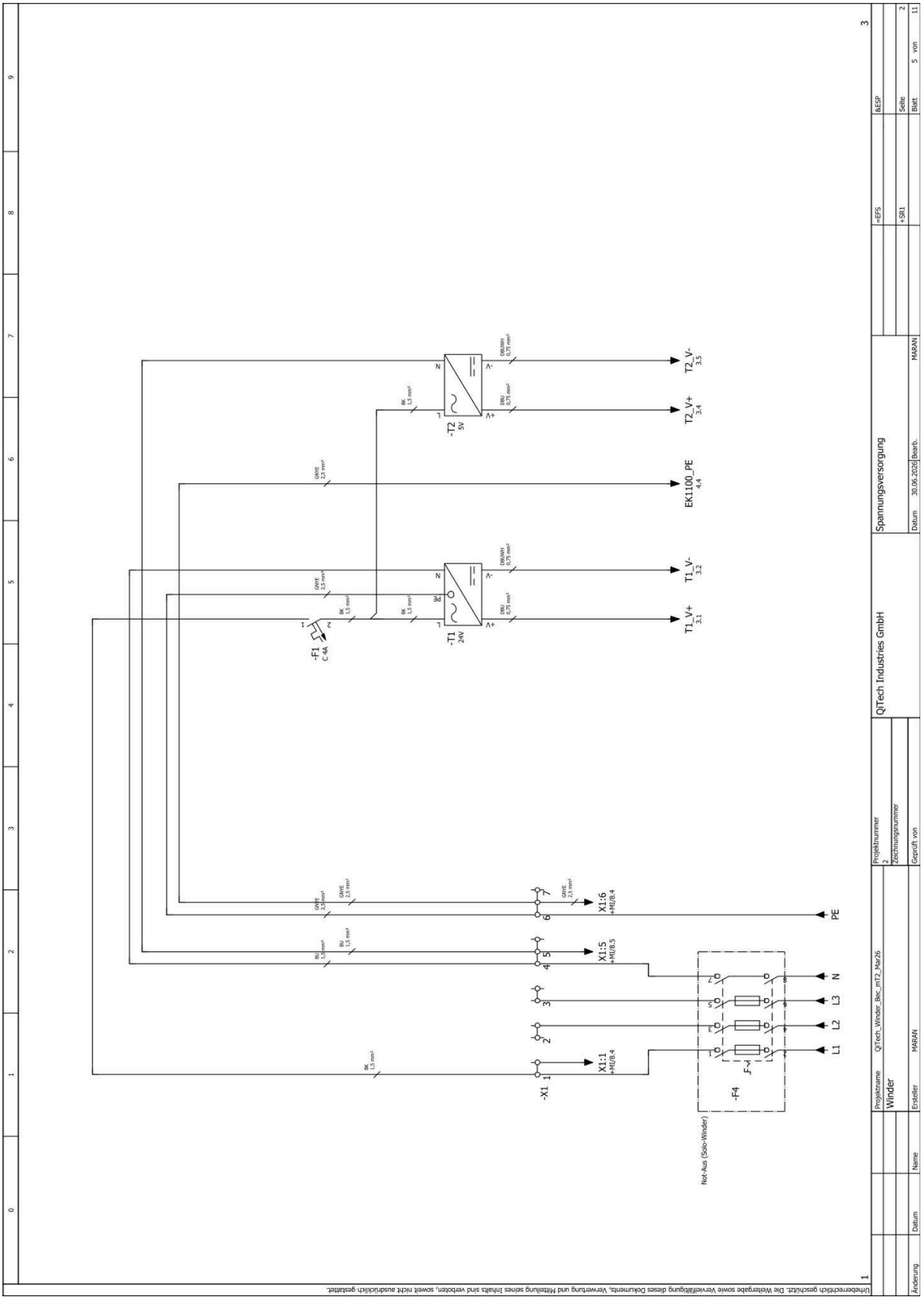
Annex C: Electrical documentation

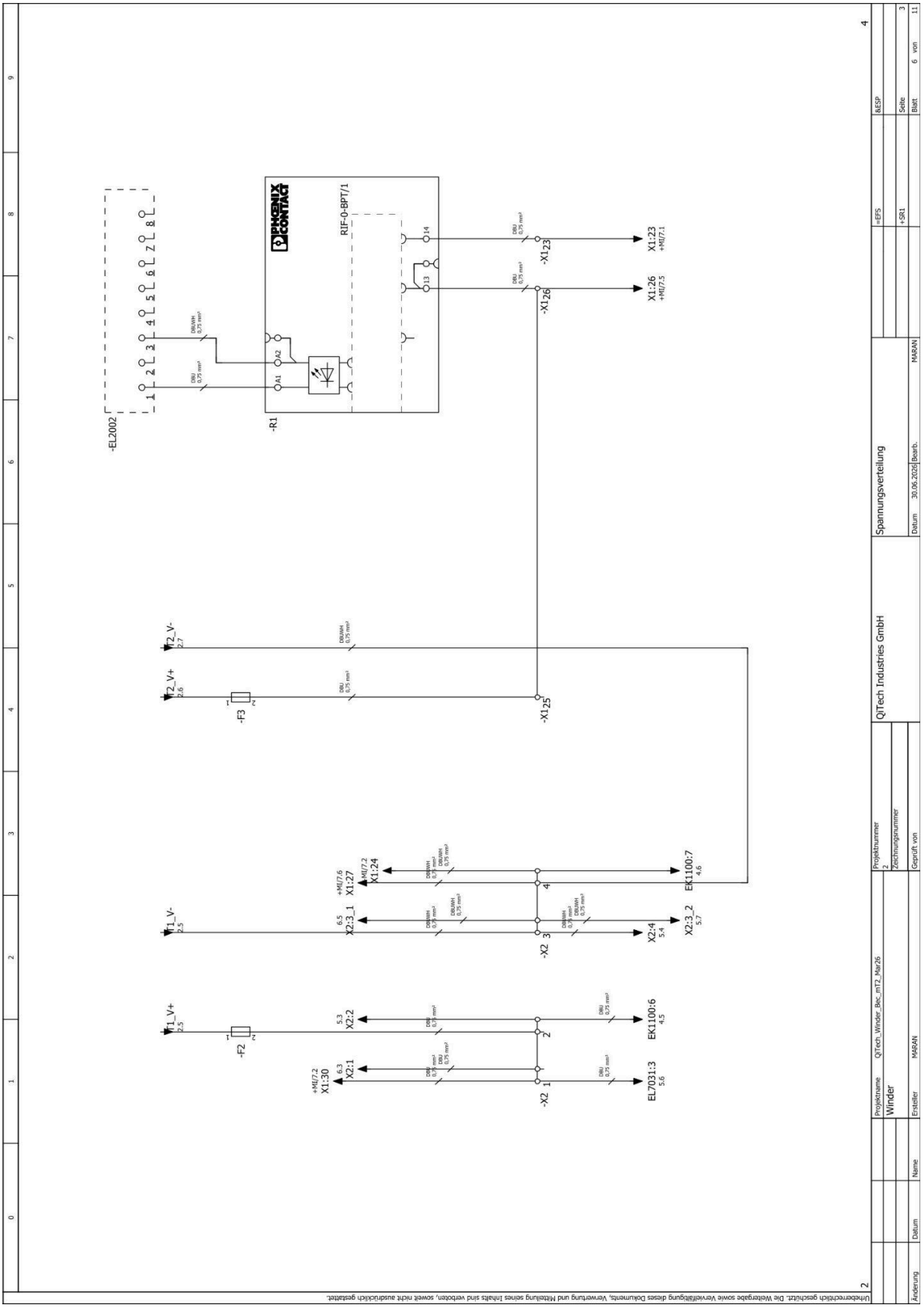


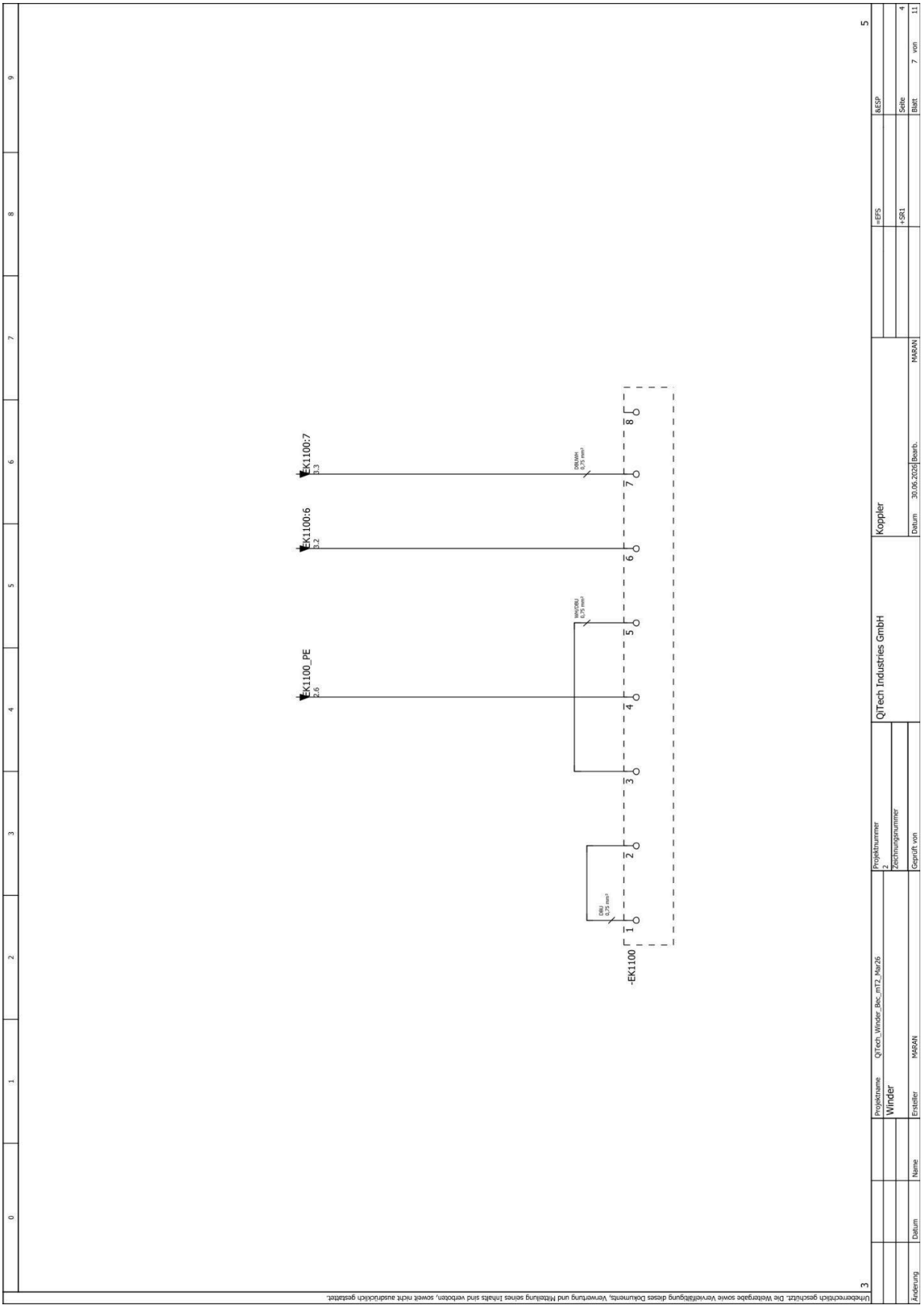

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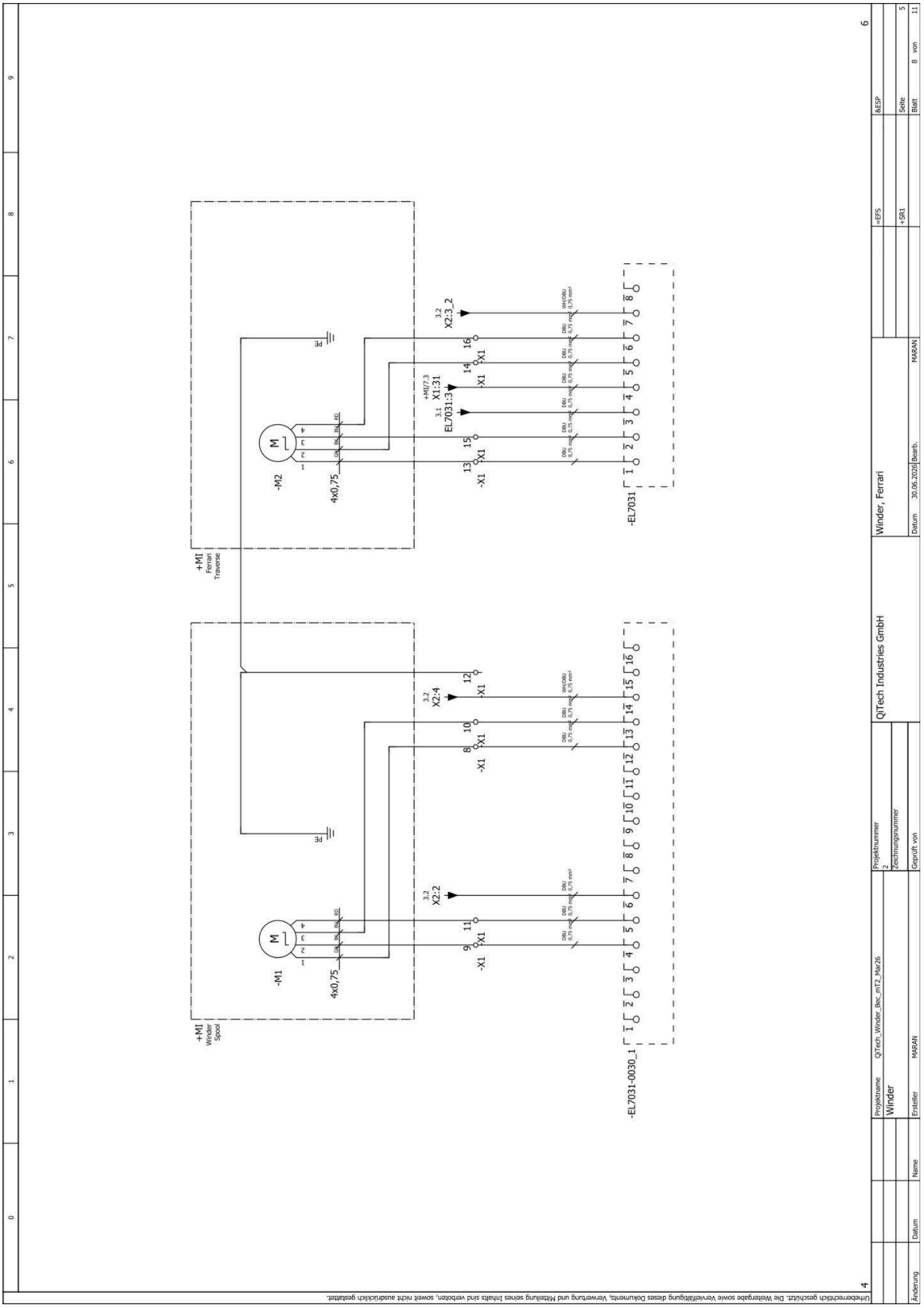

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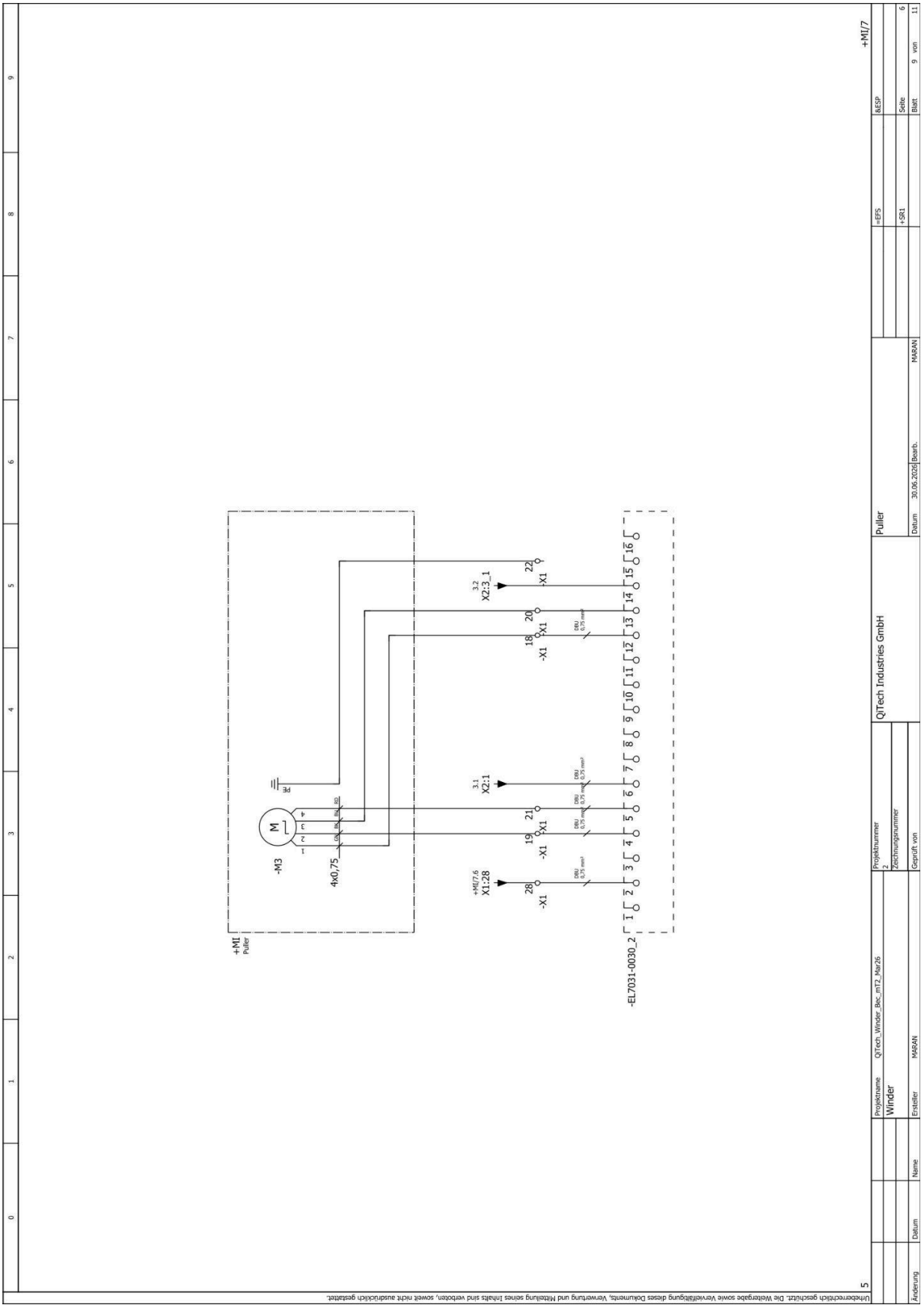

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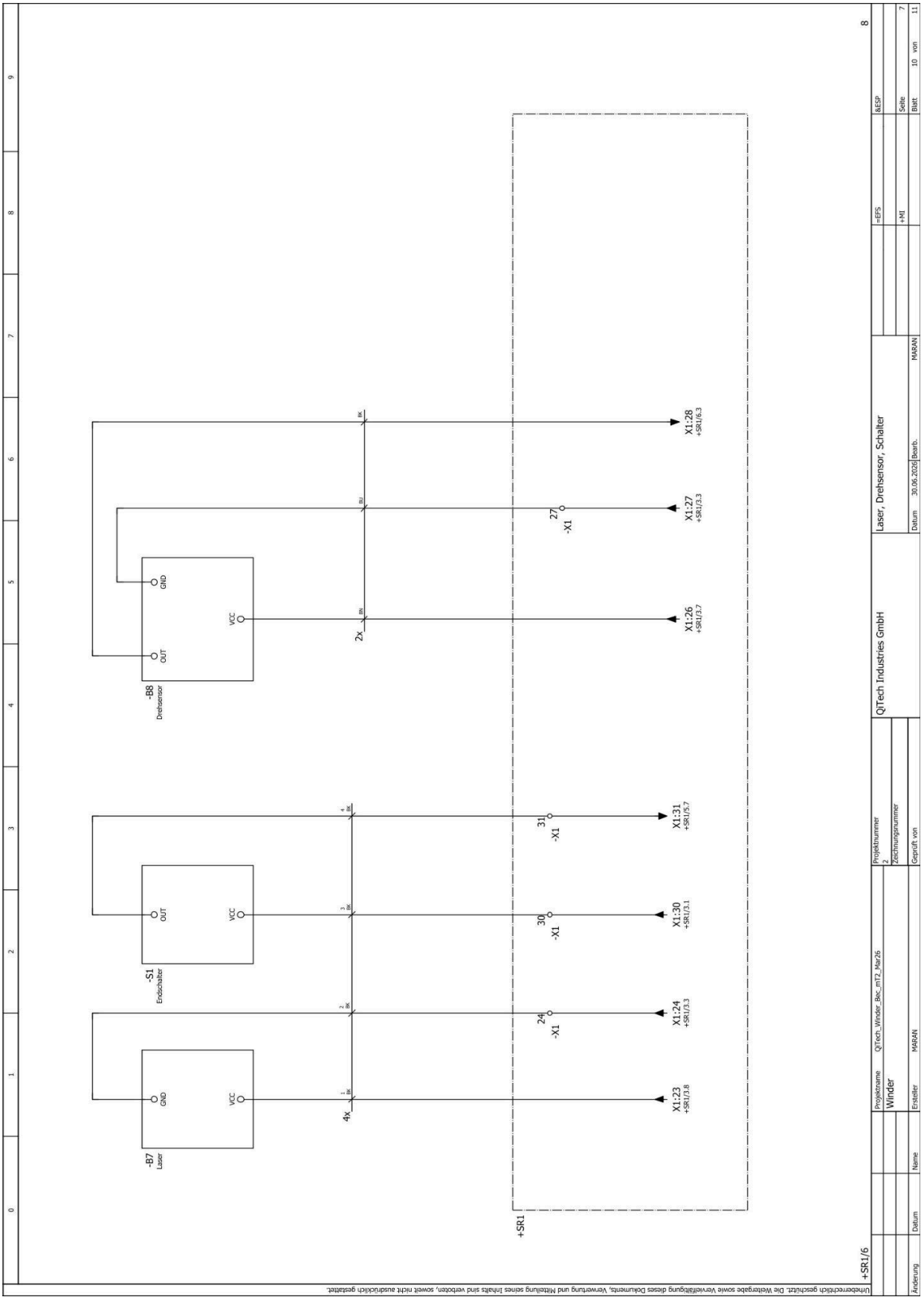

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PE
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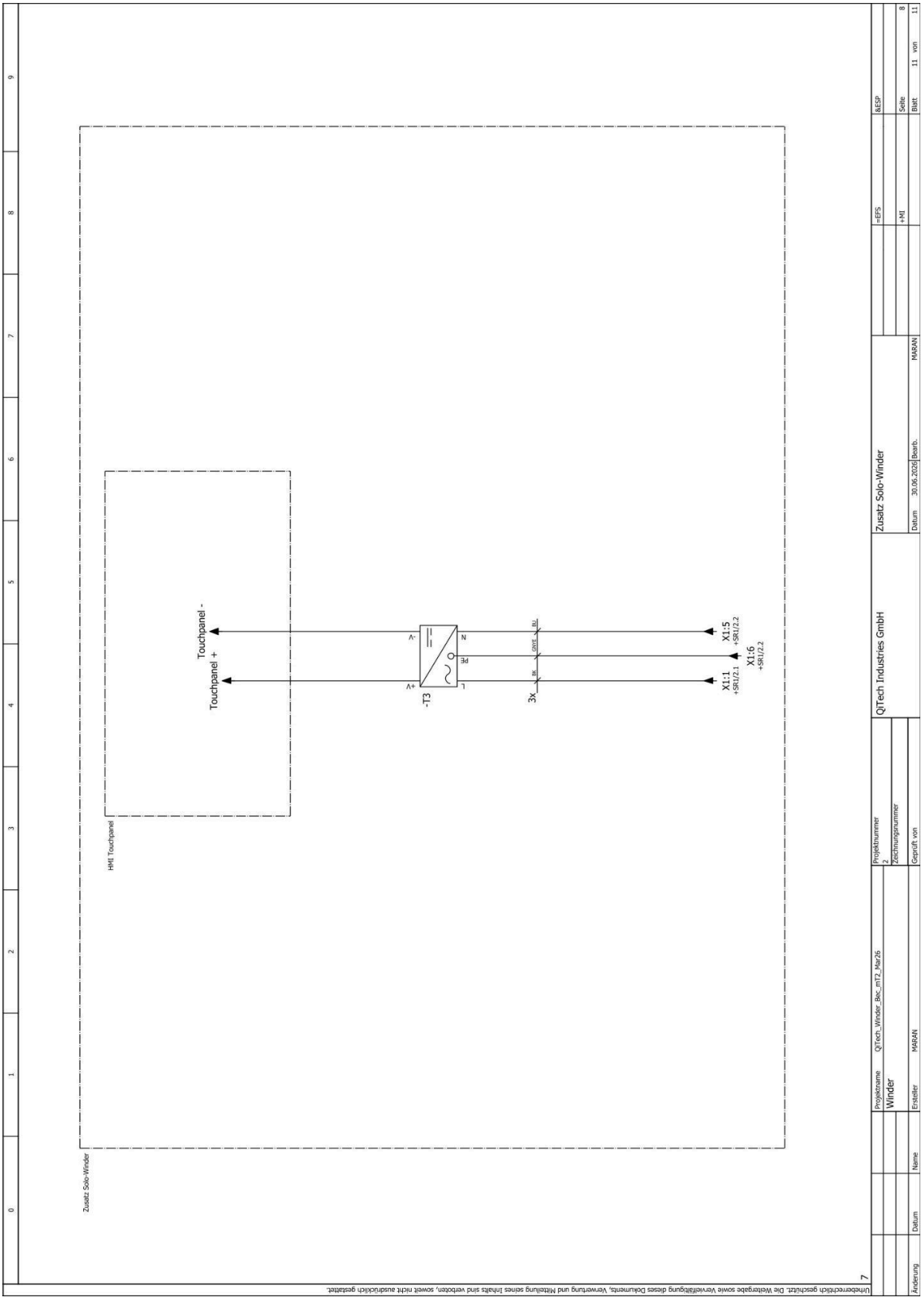

-EL2002
-R1
RIF-0-BPT/1
-X123
-X126
-X125
-F3
-F2
-X2
QiTech Industries GmbH
Spannungsverteilung
Winder
QiTech_Winder_Bec_mT2_Mar26
MARAN
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-EK1100
EK1100_PE
2.6
EK1100:6
3.2
EK1100:7
3.3
QTech Industries GmbH
Koppler
Winder
QTech_Winder_Bec_mT2_Mar26
MARAN
30.06.2026
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+MI
Winder
Spool
+MI
Ferrari
Traverse
-M1
-M2
4x0,75
PE
-EL7031-0030_1
-EL7031
-X1
QTech Industries GmbH
Winder, Ferrari
Winder
QTech_Winder_Bec_mT2_Mar26
MARAN
Datum
Name
Ersteller
Änderung
Projektname
Projektnummer
Zeichnungsnummer
Geprüft von
Bearb.
Seite
Blatt
=EFS
+SR1
&ESP
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+M1/7
Puller
QTech Industries GmbH
Projektname
Winder
Projektnummer
Zeichnungsnummer
Datum
Name
Ersteller
MARAN
Geprüft von
=EFS
+SR1
&ESP
Seite
Blatt
Änderung
-M3
4x0,75
+M1
Puller
-EL7031-0030_2
X2:1
X2:3_1
+M1/7.6
X1:28
-X1


-B7
Laser
GND
VCC
4x
-S1
Endschalter
OUT
VCC
-B8
Drehsensor
GND
OUT
VCC
2x
+SR1
X1:23
+SR1/3.8
X1:24
+SR1/3.3
X1:30
+SR1/3.1
X1:31
+SR1/5.7
X1:26
+SR1/3.7
X1:27
+SR1/3.3
X1:28
+SR1/6.3
24
-X1
30
-X1
31
-X1
27
-X1
Laser, Drehsensor, Schalter
QiTech Industries GmbH
Projektname
Winder
QiTech_Winder_Bkc_m72_Mar26
Zeichnungsnummer
2
Geprüft von
Ersteller
MARAN
Name
Datum
Änderung
Datum
30.06.2026
Bearb.
MARAN
=EFS
+MI
&ESP
Seite
7
Blatt
10 von 11
+SR1/6
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Zusatz Solo-Winder
HMI Touchpanel
Touchpanel -
Touchpanel +
-T3
~V
+V
N
PE
L
GNYE
BK
3x
X1:1
+SR1/2.1
X1:5
+SR1/2.2
X1:6
+SR1/2.2
QITech Industries GmbH
Zusatz Solo-Winder
Datum 30.06.2026
Bearb. MARAN
=EFS
+MI
&ESP
Seite 8
Blatt 11 von 11
Projektname Winder
QITech_Winder_Bec_m72_Mar26
Projektnummer 2
Zeichnungsnummer
Ersteller MARAN
Geprüft von
Änderung
Datum
Name
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